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MCAL27 - NETWORKING WITH LINUX LAB 2024- 2025

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PRACTICAL NO. 1

Installing NS-3 in Ubuntu

Network simulator is a tool used for simulating the real world network on one computer by writing scripts in C++ or Python. Normally if we want to perform experiments, to see how our network works using various parameters. We don't have required number of computers and routers for making different topologies. Even if we have these resources it is very expensive to build such a network for experiment purposes.

So to overcome these drawbacks we used NS3, which is a discrete event network simulator for Internet. NS3 helps to create various virtual nodes (i.e., computers in real life) and with the help of various Helper classes it allows us to install devices, internet stacks, application, etc to our nodes.

Using NS3 we can create PointToPoint, Wireless, CSMA, etc connections between nodes. PointToPoint connection is same as a LAN connected between two computers. Wireless connection is same as WiFi connection between various computers and routers. CSMA connection is same as bus topology between computers. After building connections we try to install NIC to every node to enable network connectivity.

When network cards are enabled in the devices, we add different parameters in the channels (i.e., real world path used to send data) which are data-rate, packet size, etc. Now we use Application to generate traffic and send the packets using these applications.

Steps for installing NS-3 in Ubuntu

Step 1: Update the system

\$ [sudo apt update](#)

Step 2: Prerequisites for installing NS

[\\$sudo apt install build-essential autoconf automake libxmu-dev g++ python3 python3-dev pkg-config sqlite3 cmake python3-setuptools git qtbase5-dev qtchooser qt5-qmake qtbase5-dev-tools gir1.2-gobject-2.0 python3-gi python3-gi-cairo python3-pygraphviz gir1.2-gtk-3.0 ipython3 openmpi-bin openmpi-common openmpi-doc libopenmpi-dev autoconf cvs bzip2 unrar gsl-bin libgsl-dev libgslcblas0 wireshark tcpdump sqlite3 libsqlite3-dev libxml2 libxml2-dev libc6-dev libc6-dev-i386 libclang-dev llvm-dev automake python3-pip libxml2 libxml2-dev libboost-all-dev](#)

Step 3 : Now download the ns3 3.35 from <https://nsnam.org>

Step 4 : Copy the softwares from the Downloads/ folder to the home folder (in my case its /home/ns-3/)

Now extract both the versions using the GUI method. Just right click and click "Extract Here"

Now we will install ns-3.35

Step 5 :

```
$ cd
```

```
$ cd ns-allinone-3.35/
```

```
$ ./build.py --enable-examples --enable-tests
```

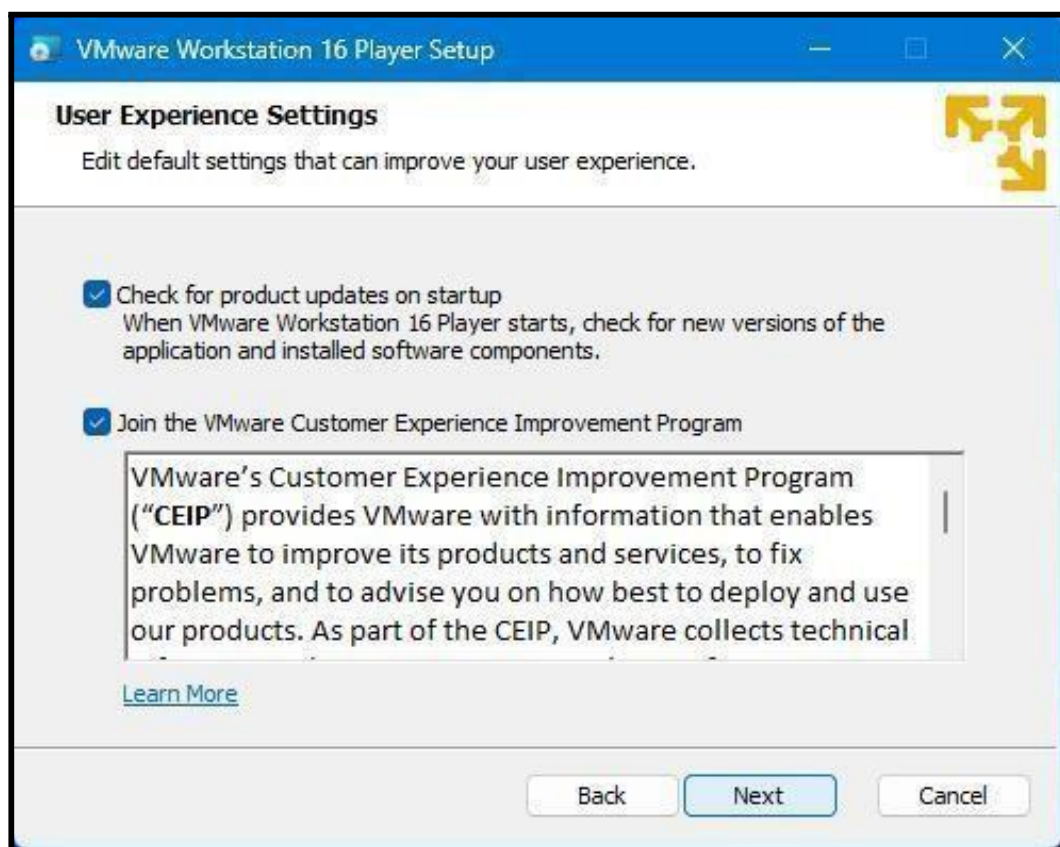
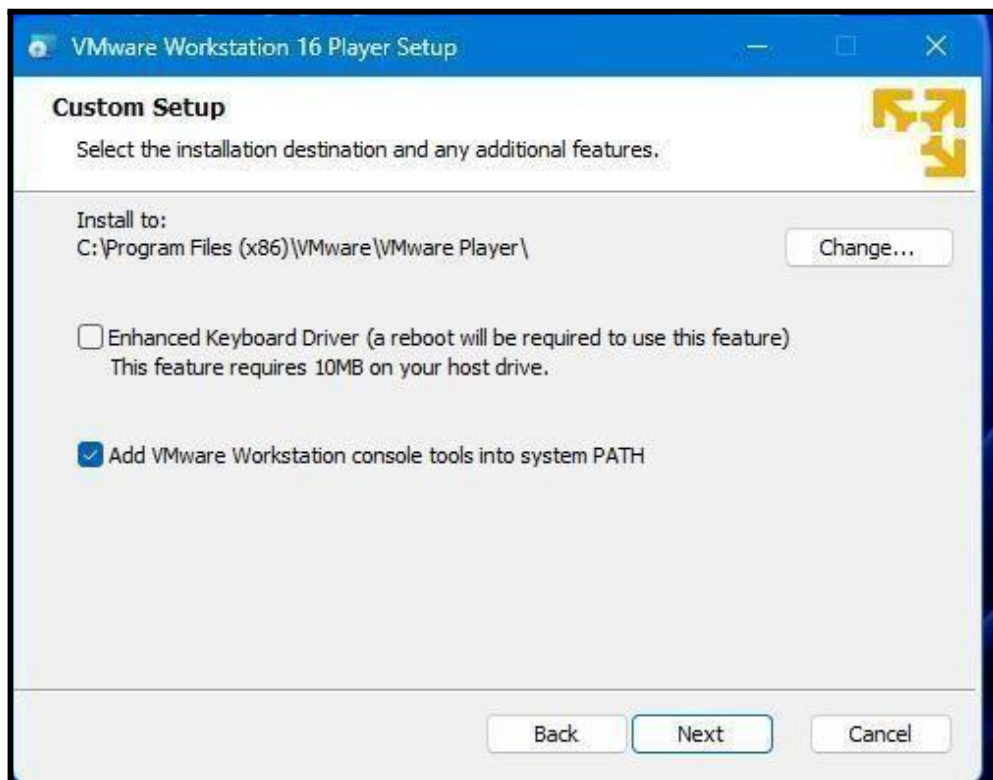
In case, if you get the following error pybindgen(ns3 module antenna) Do this step and repeat the above step





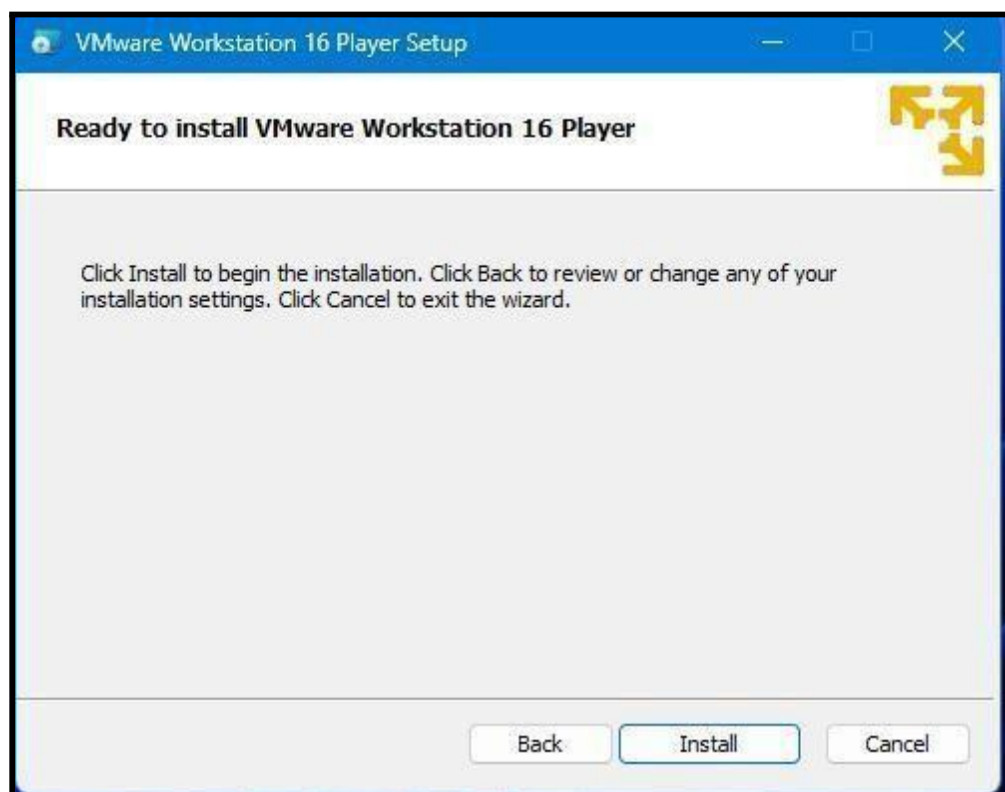
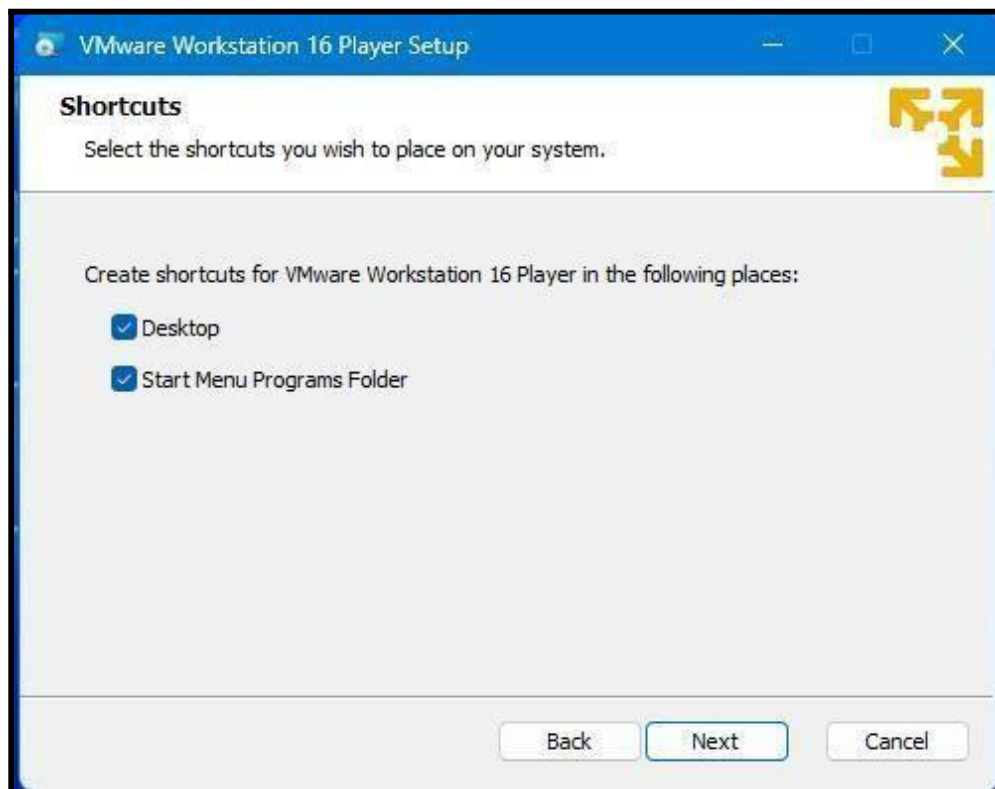
Create a New virtual Machine in VMWare.

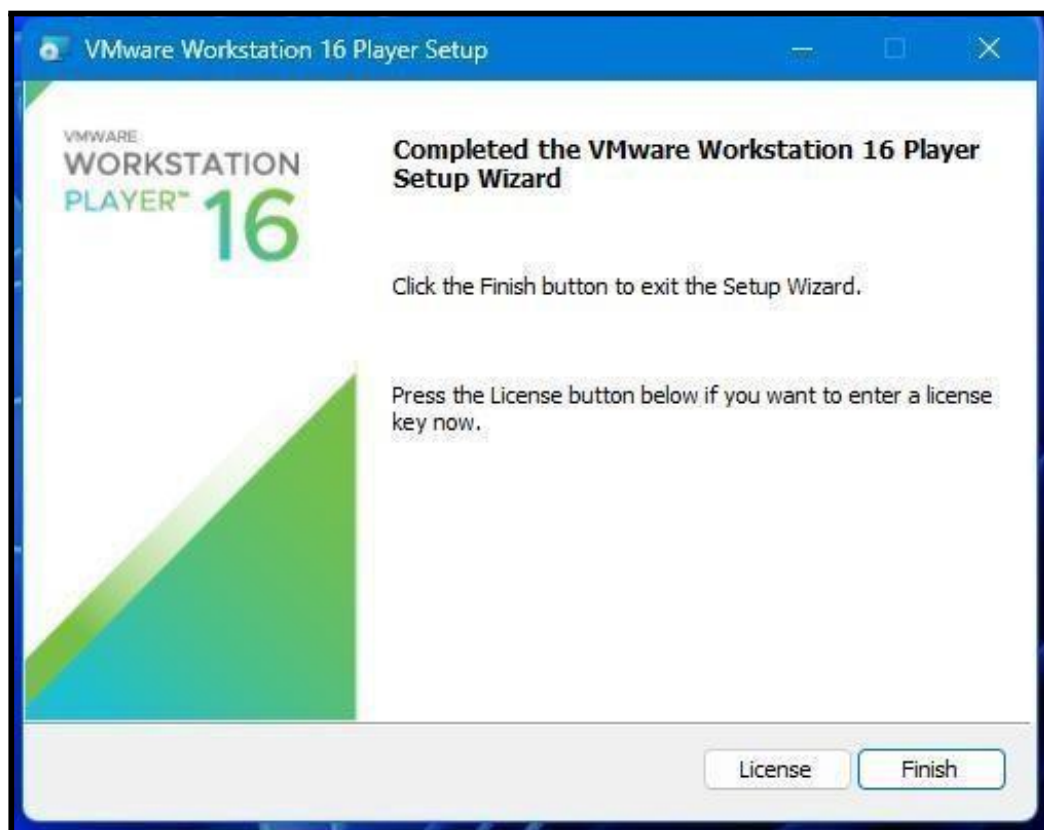
Screenshot shows the process of creating a new virtual machine in VMware for running Ubuntu.



Install the necessary Libraries like NS3, Netanime etc.

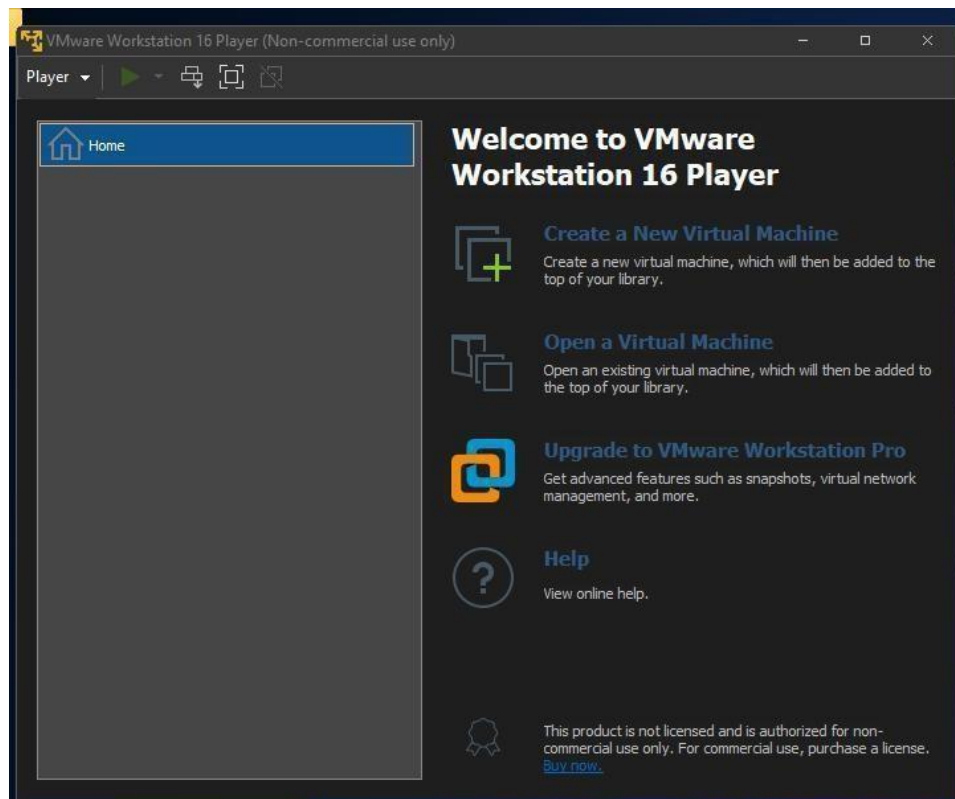
Screenshot displays installation of NS3 and NetAnim libraries through terminal commands in Ubuntu.

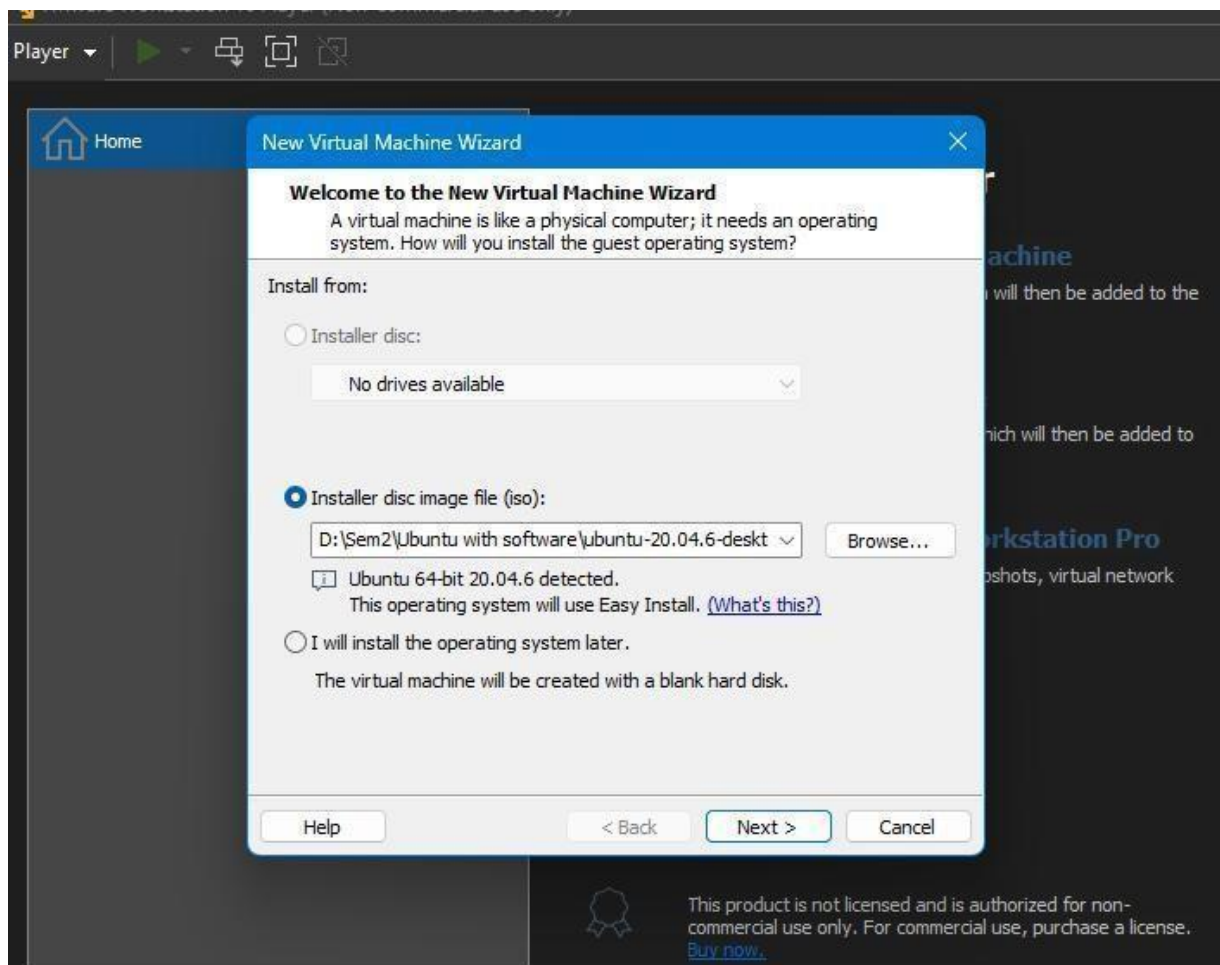


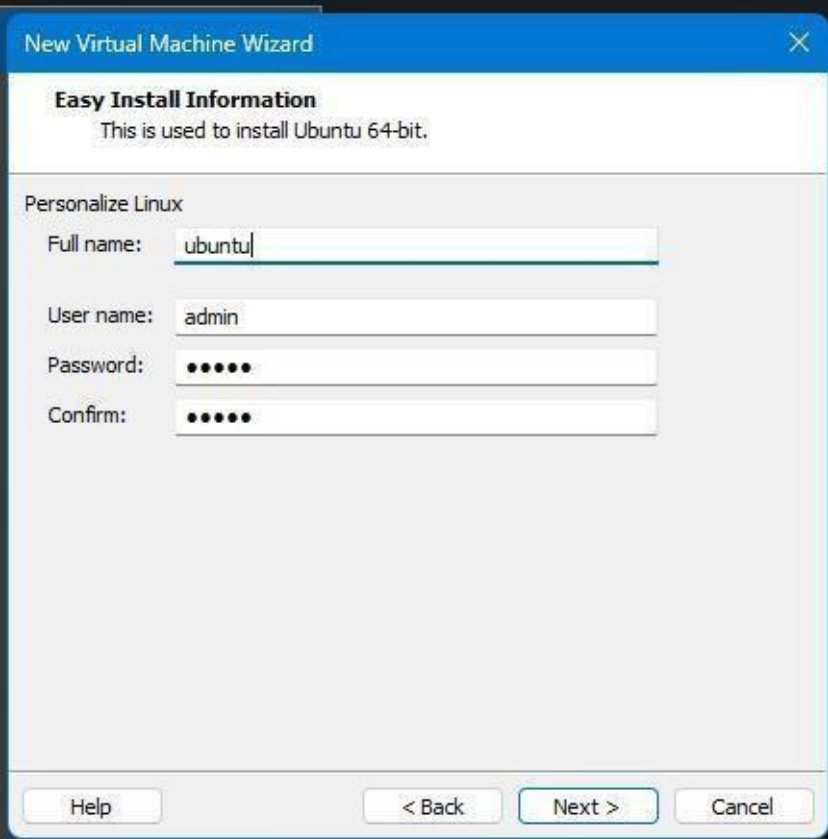




Create a New virtual Machine in VMWare







New Virtual Machine Wizard

Easy Install Information
This is used to install Ubuntu 64-bit.

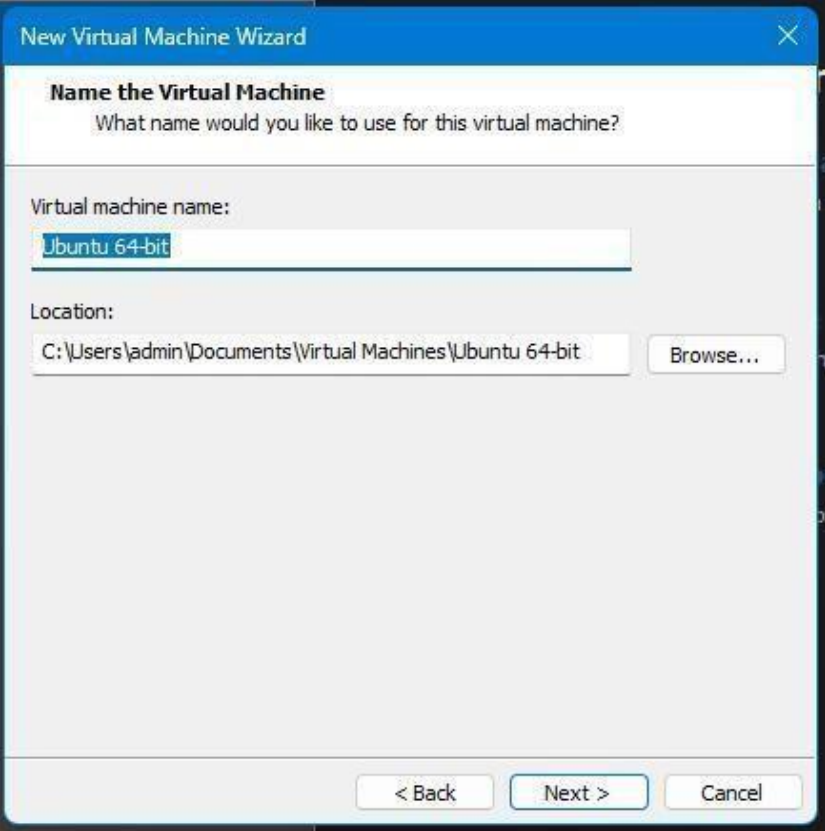
Personalize Linux

Full name:

User name:

Password:

Confirm:

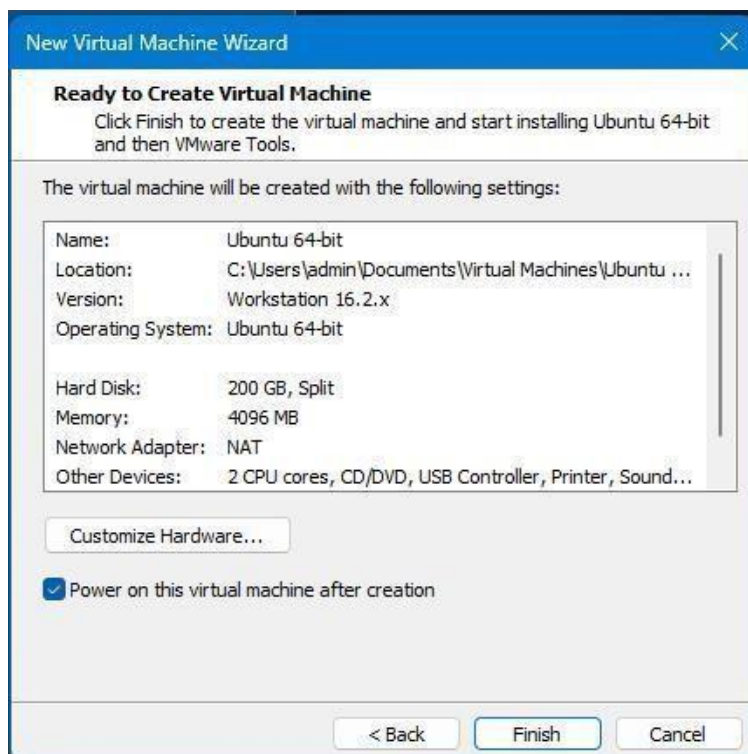
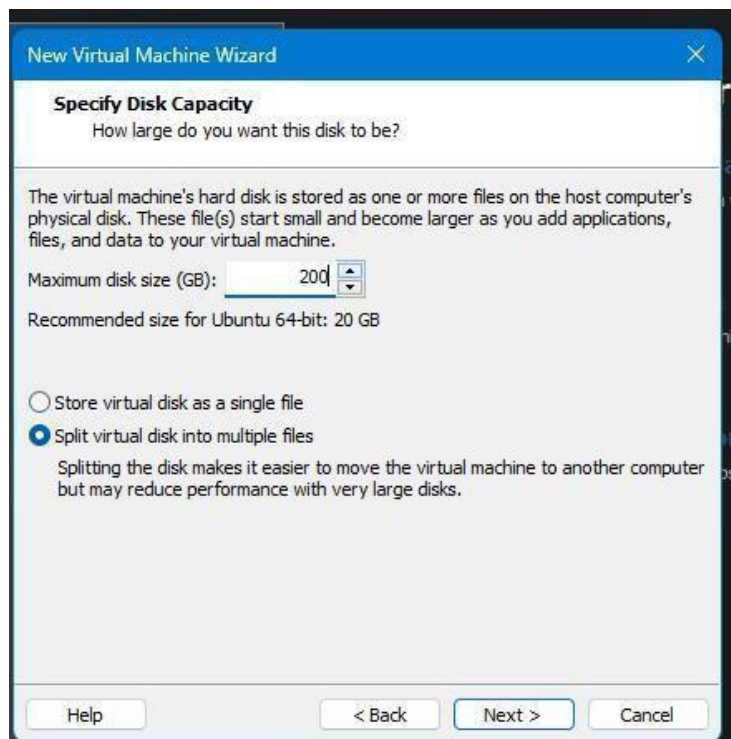


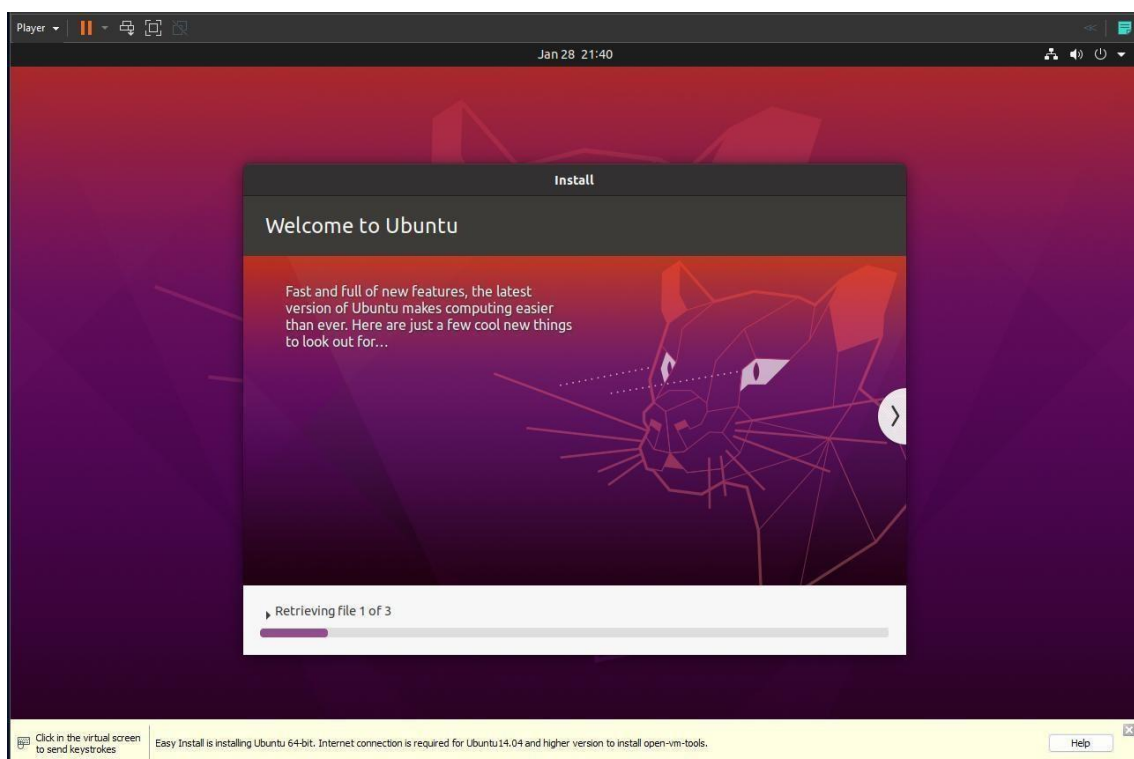
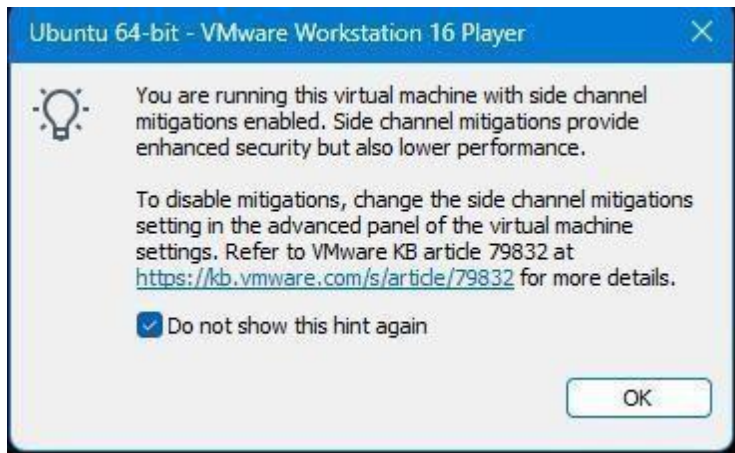
New Virtual Machine Wizard

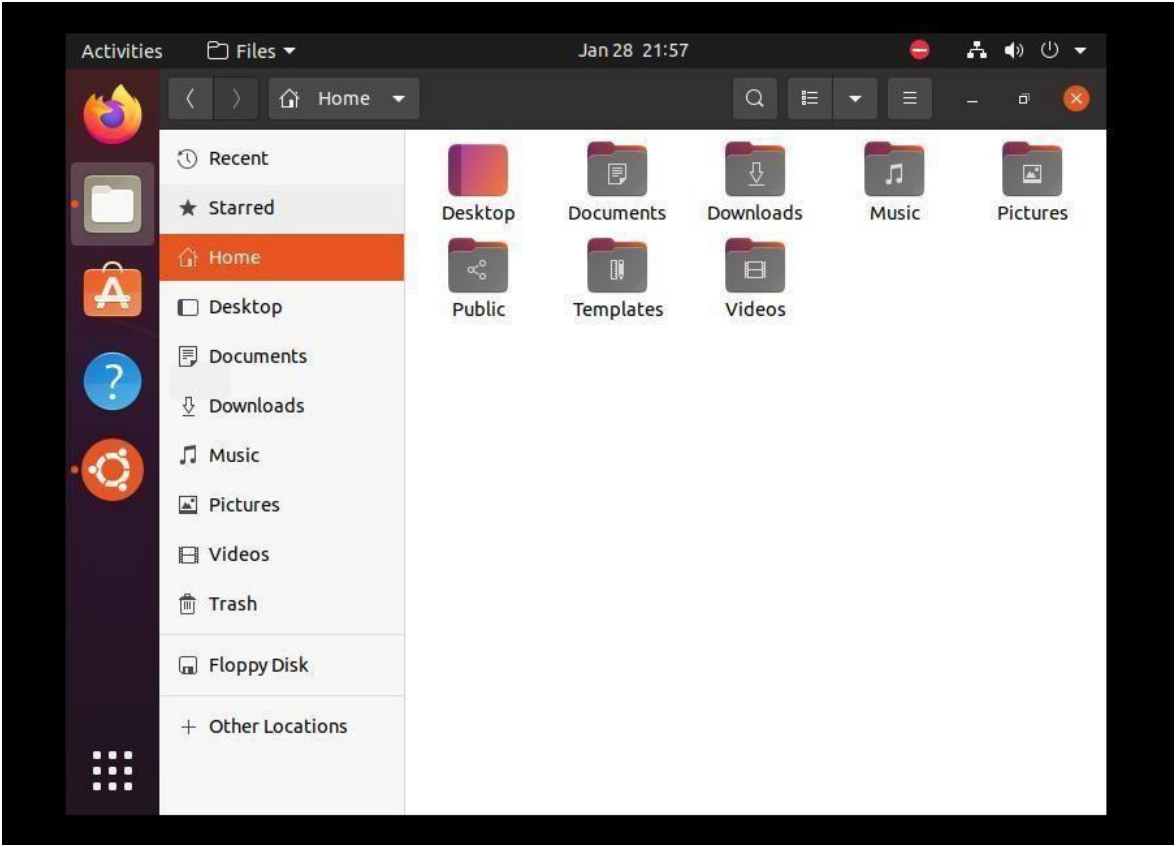
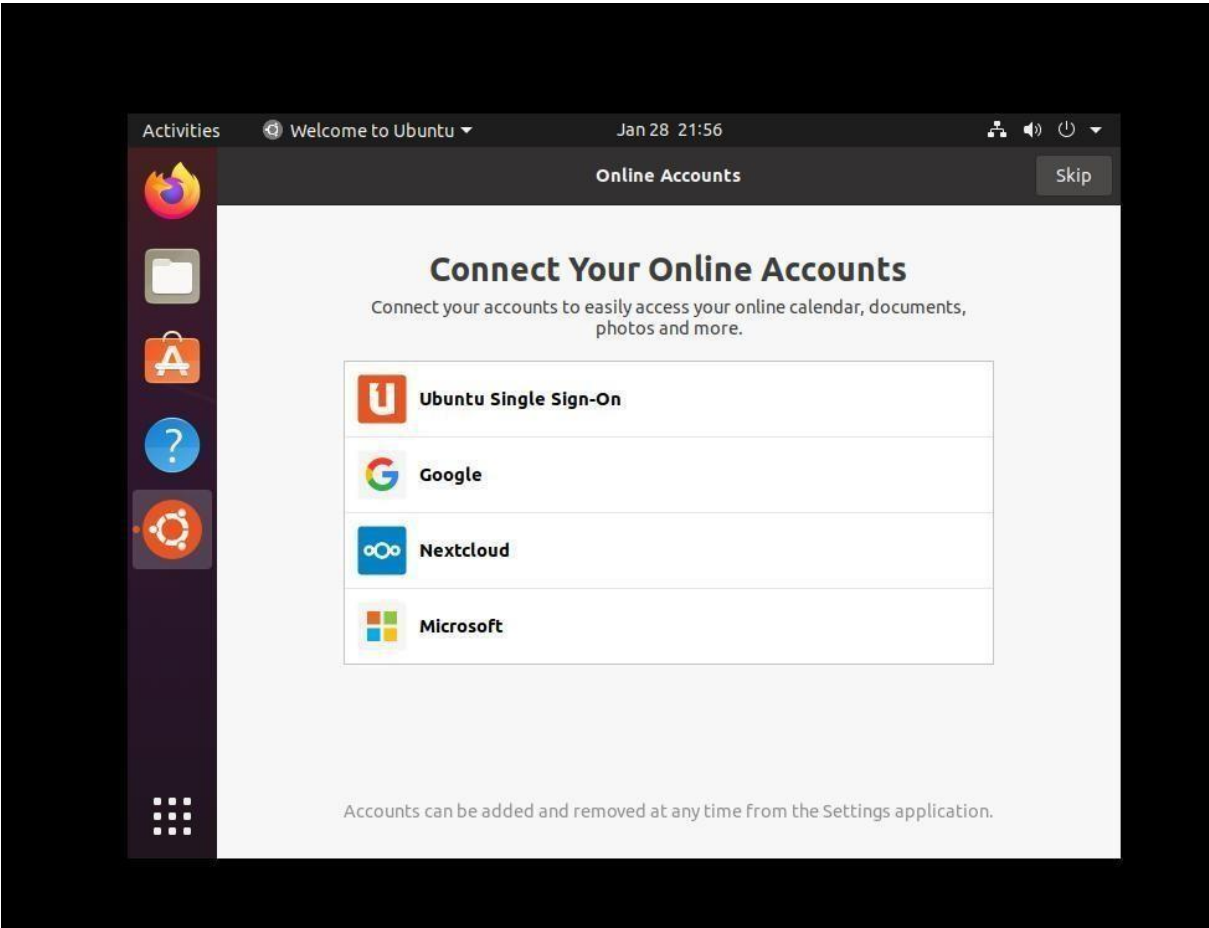
Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

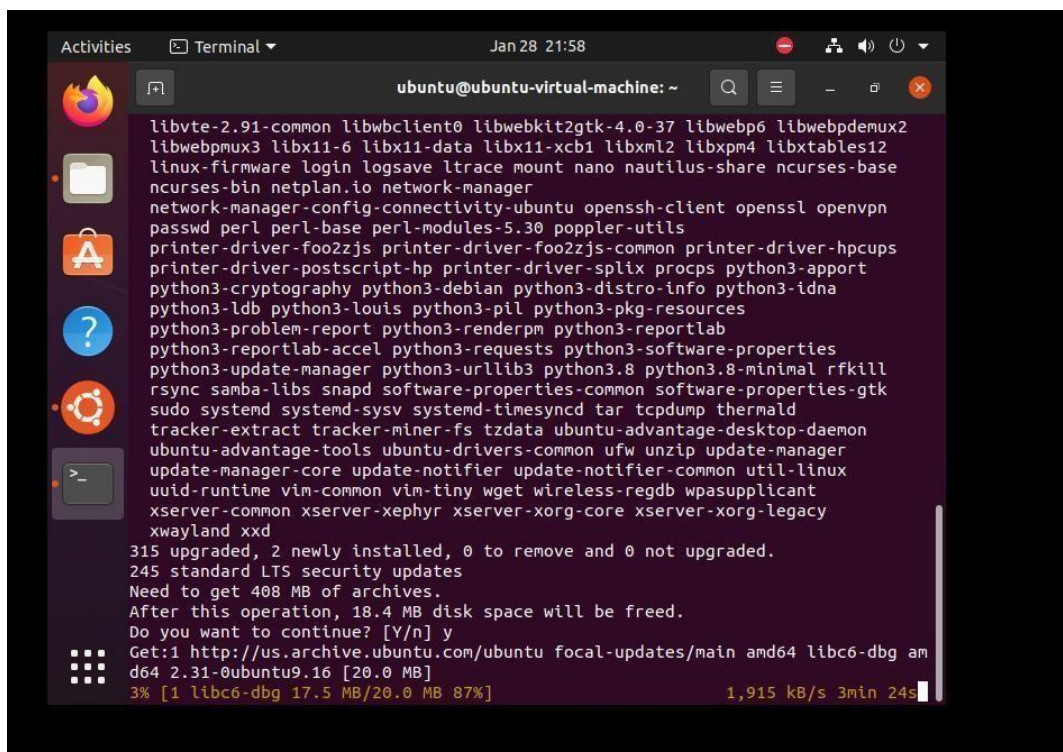
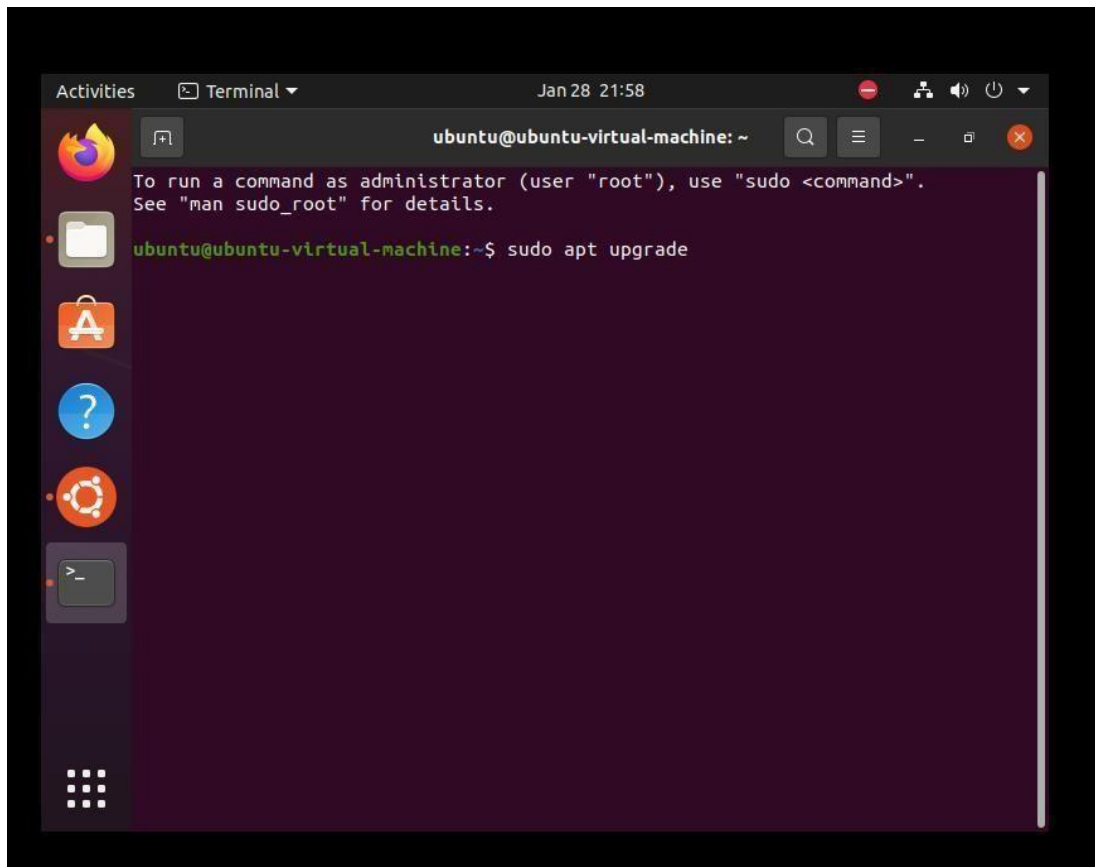
Location:

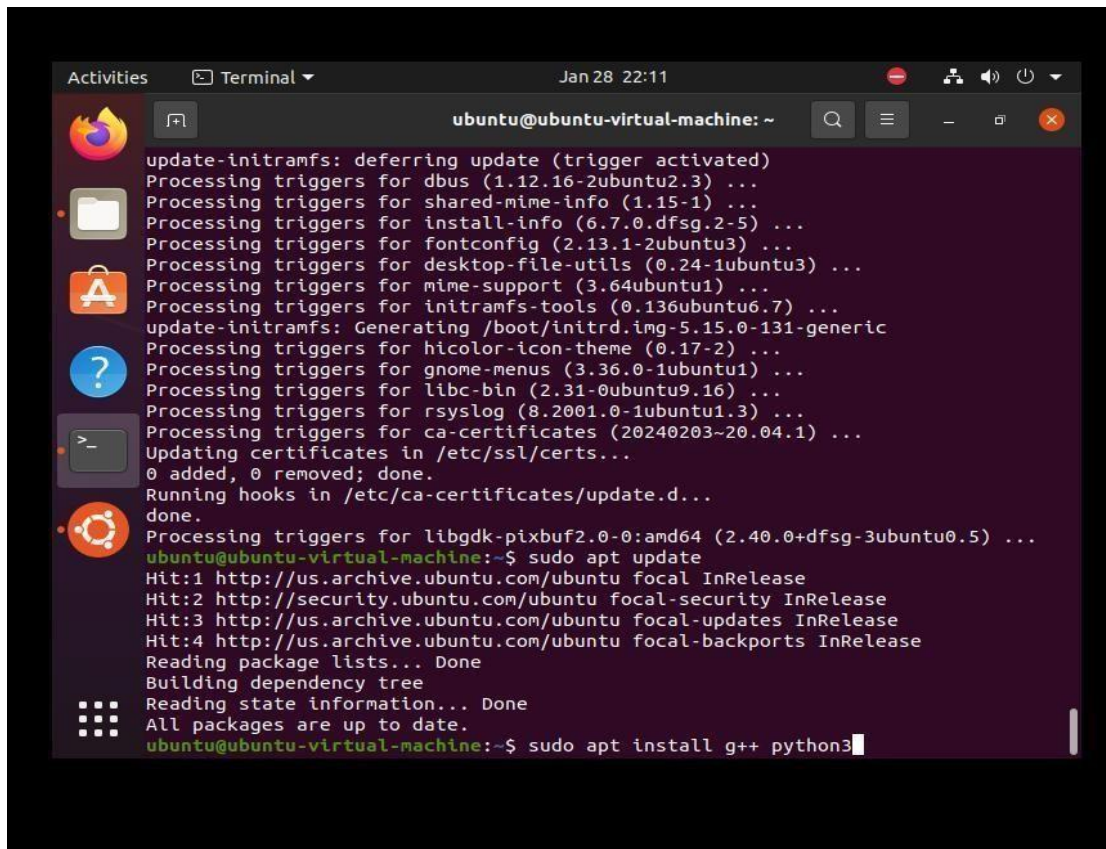






Install the necessary Libraries like NS3, Netanime etc.





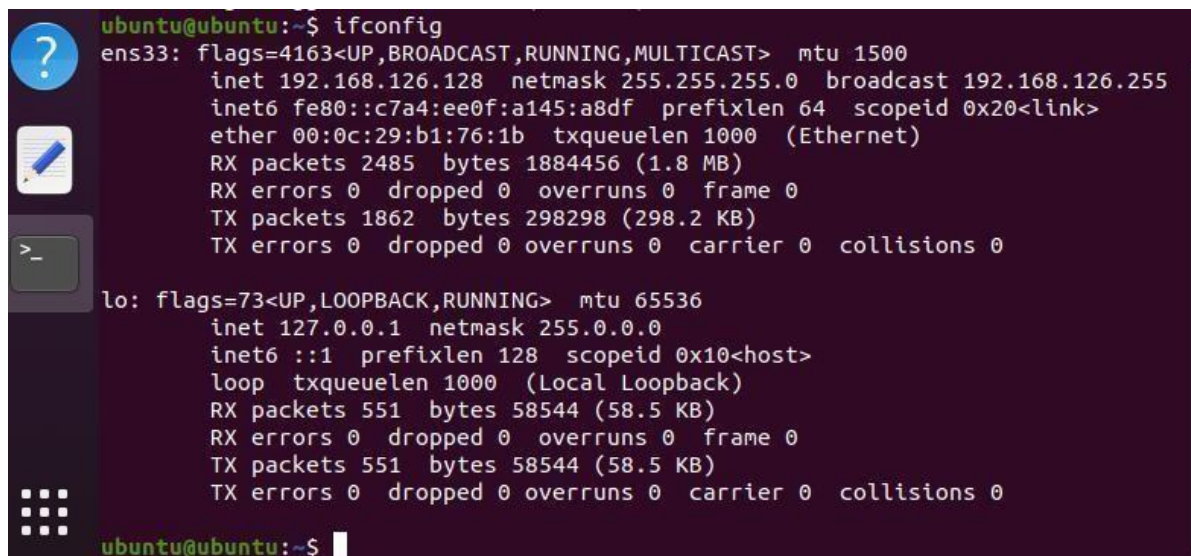
The screenshot shows a terminal window titled 'Terminal' with the date and time 'Jan 28 22:11'. The prompt is 'ubuntu@ubuntu-virtual-machine: ~'. The terminal output displays the following:

```
update-initramfs: deferring update (trigger activated)
Processing triggers for dbus (1.12.16-2ubuntu2.3) ...
Processing triggers for shared-mime-info (1.15-1) ...
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for fontconfig (2.13.1-2ubuntu3) ...
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: Generating /boot/initrd.img-5.15.0-131-generic
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.16) ...
Processing triggers for rsyslog (8.2001.0-1ubuntu1.3) ...
Processing triggers for ca-certificates (20240203~20.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for libgdk-pixbuf2.0-0:amd64 (2.40.0+dfsg-3ubuntu0.5) ...
ubuntu@ubuntu-virtual-machine:~$ sudo apt update
Hit:1 http://us.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
ubuntu@ubuntu-virtual-machine:~$ sudo apt install g++ python3
```

PRACTICAL NO. 2

Linux network commands – ifconfig, ip, ping, netstat, traceroute, nslookup, route, hostname.

Ifconfig: used for **configuring and displaying network interfaces.**



```
ubuntu@ubuntu:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.126.128 netmask 255.255.255.0 broadcast 192.168.126.255
    inet6 fe80::c7a4:ee0f:a145:a8df prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:b1:76:1b txqueuelen 1000 (Ethernet)
    RX packets 2485 bytes 1884456 (1.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1862 bytes 298298 (298.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 551 bytes 58544 (58.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 551 bytes 58544 (58.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ubuntu@ubuntu:~$
```

ip : Displays and manipulates routing, devices and tunnels.



```
ubuntu@ubuntu:~$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { link | address | addrlabel | route | rule | neigh | ntable |
                  tunnel | tuntap | maddress | mroute | mrule | monitor | xfrm
                  |
                  netns | l2tp | fou | macsec | tcp_metrics | token | netconf
                  |
                  | ila |
                  vrf | sr | nexthop }
OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
             -h[uman-readable] | -iec | -j[son] | -p[retty] |
             -f[amily] { inet | inet6 | mpls | bridge | link } |
             -4 | -6 | -I | -D | -M | -B | -O |
             -l[oops] { maximum-addr-flush-attempts } | -br[ief] |
             -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename]
             |
             -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
             -c[olor]}
```

Ip -v : Shows version and output of ip commands

```
ubuntu@ubuntu:~$ ip -V
ip utility, iproute2-ss200127
ubuntu@ubuntu:~$
```

Ip addr : Shows ip addresses

```
ubuntu@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b1:76:1b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.126.128/24 brd 192.168.126.255 scope global dynamic noprefixroute ens33
        valid_lft 1537sec preferred_lft 1537sec
    inet6 fe80::c7a4:ee0f:a145:a8df/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

Ip addr show : Displays detailed ip configuration

```
ubuntu@ubuntu:~$ ip addr show lo
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

Ip link : Show and configures network interfaces

```
ubuntu@ubuntu:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:b1:76:1b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
```

Ip route show : Displays the current routing table


```
ubuntu@ubuntu:~$ ip route show
default via 192.168.126.2 dev ens33 proto dhcp metric 100
169.254.0.0/16 dev ens33 scope link metric 1000
192.168.126.0/24 dev ens33 proto kernel scope link src 192.168.126.128 metric 100
ubuntu@ubuntu:~$
```

Ifconfig -s : Shows a short summary of network interfaces

```
ubuntu@ubuntu:~$ ifconfig -s
Iface      MTU     RX-OK RX-ERR RX-DRP RX-OVR      TX-OK TX-ERR TX-DRP TX-OVR Flg
ens33      1500    6189   0      0 0        3081   0      0  0 BMRU
lo         65536    605   0      0 0          605   0      0  0 LRU
ubuntu@ubuntu:~$
```

Dig : Queries DNS Servers for information.

```
ubuntu@ubuntu:~$ dig google.com

; <<>> DiG 9.18.30-0ubuntu0.20.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62423
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
Text Editor
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 5       IN      A      142.251.42.110

;; Query time: 3 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Feb 17 22:21:22 PST 2025
;; MSG SIZE rcvd: 55
```

Nslookup : Queries internet name servers interactively for domain name .

```
ubuntu@ubuntu:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.183.110
Name:   google.com
Address: 2404:6800:4009:823::200e
```

Netstat -at : Displays all active TCP Connections

```
ubuntu@ubuntu:~$ netstat -at
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:domain        0.0.0.0:*               LISTEN
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 ubuntu:47758            93.243.107.34.bc.:https ESTABLISHED
tcp        0      0 ubuntu:47750            93.243.107.34.bc.:https ESTABLISHED
tcp        0      0 ubuntu:46064            209.100.149.34.bc:https ESTABLISHED
tcp6       0      0 ip6-localhost:ipp      [::]:*                  LISTEN
ubuntu@ubuntu:~$
```

Host: performs DNS lookup to find ip addresses

```
ubuntu@ubuntu:~$ host google.com
google.com has address 142.251.42.110
google.com has IPv6 address 2404:6800:4009:832::200e
google.com mail is handled by 10 smtp.google.com.
ubuntu@ubuntu:~$
```

Ping : sends ICMP echo request to test network

```

ubuntu@ubuntu:~$ ping -c 5 google.com
PING google.com (142.250.183.110) 56(84) bytes of data.
64 bytes from bom12s13-in-f14.1e100.net (142.250.183.110): icmp_seq=1 ttl=128 t
ime=2.13 ms
64 bytes from bom12s13-in-f14.1e100.net (142.250.183.110): icmp_seq=2 ttl=128 t
ime=2.02 ms
64 bytes from bom12s13-in-f14.1e100.net (142.250.183.110): icmp_seq=3 ttl=128 t
ime=2.14 ms
64 bytes from bom12s13-in-f14.1e100.net (142.250.183.110): icmp_seq=4 ttl=128 t
ime=2.23 ms
64 bytes from bom12s13-in-f14.1e100.net (142.250.183.110): icmp_seq=5 ttl=128 t
ime=2.11 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 2.018/2.126/2.231/0.068 ms
ubuntu@ubuntu:~$

```

Traceroute : Tracethe route packets take to reach a network host

```

ubuntu@ubuntu:~$ sudo traceroute -T 184.95.56.34
traceroute to 184.95.56.34 (184.95.56.34), 30 hops max, 60 byte packets
 1 _gateway (192.168.126.2) 0.092 ms 0.051 ms 0.081 ms
 2 speedtest.phoenixnap.com (184.95.56.34) 0.523 ms 0.503 ms 0.740 ms
ubuntu@ubuntu:~$ sudo traceroute -T 127.0.0.53
traceroute to 127.0.0.53 (127.0.0.53), 30 hops max, 60 byte packets
 1 localhost (127.0.0.53) 0.020 ms 0.004 ms 0.004 ms
ubuntu@ubuntu:~$

```

Route : Displays or modifies the ip routing table

```

ubuntu@ubuntu:~$ route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
default _gateway 0.0.0.0 UG 100 0 0 ens33
link-local 0.0.0.0 255.255.0.0 U 1000 0 0 ens33
192.168.126.0 0.0.0.0 255.255.255.0 U 100 0 0 ens33

```

Arp : Displays or modifies the ARP

```

ubuntu@ubuntu:~$ arp
Address HWtype HWaddress Flags Mask Iface
e
192.168.126.254 ether 00:50:56:e2:86:ba C ens3
3
_gateway ether 00:50:56:eb:a2:d7 C ens3
3
ubuntu@ubuntu:~$

```

Iwconfig : Configures wireless network interfaces


```
ubuntu@ubuntu:~$ iwconfig
lo          no wireless extensions.

ens33      no wireless extensions.

ubuntu@ubuntu:~$
```

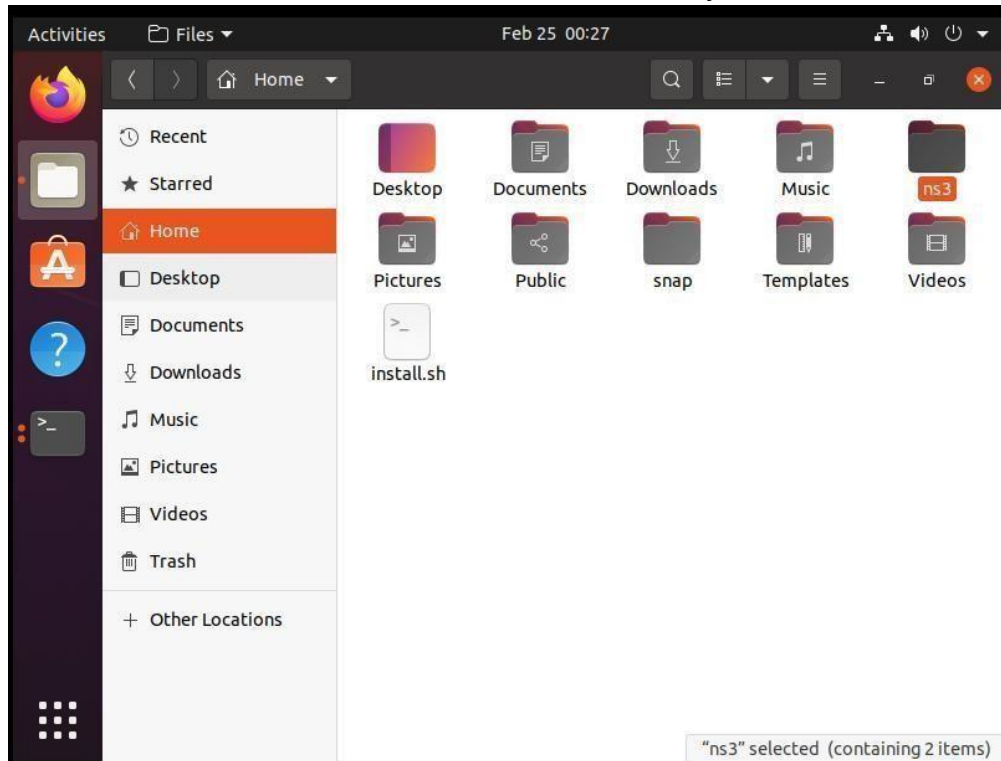
Whois : Fetches domain registration information.

```
ubuntu@ubuntu:~$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
```

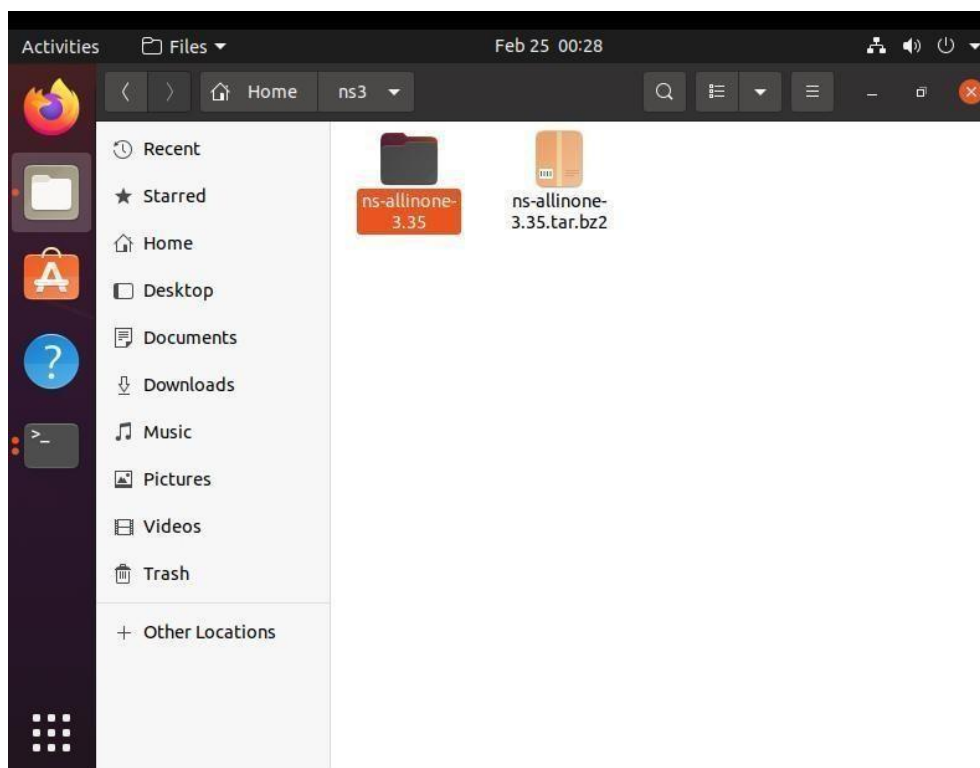

PRACTICAL NO. 3

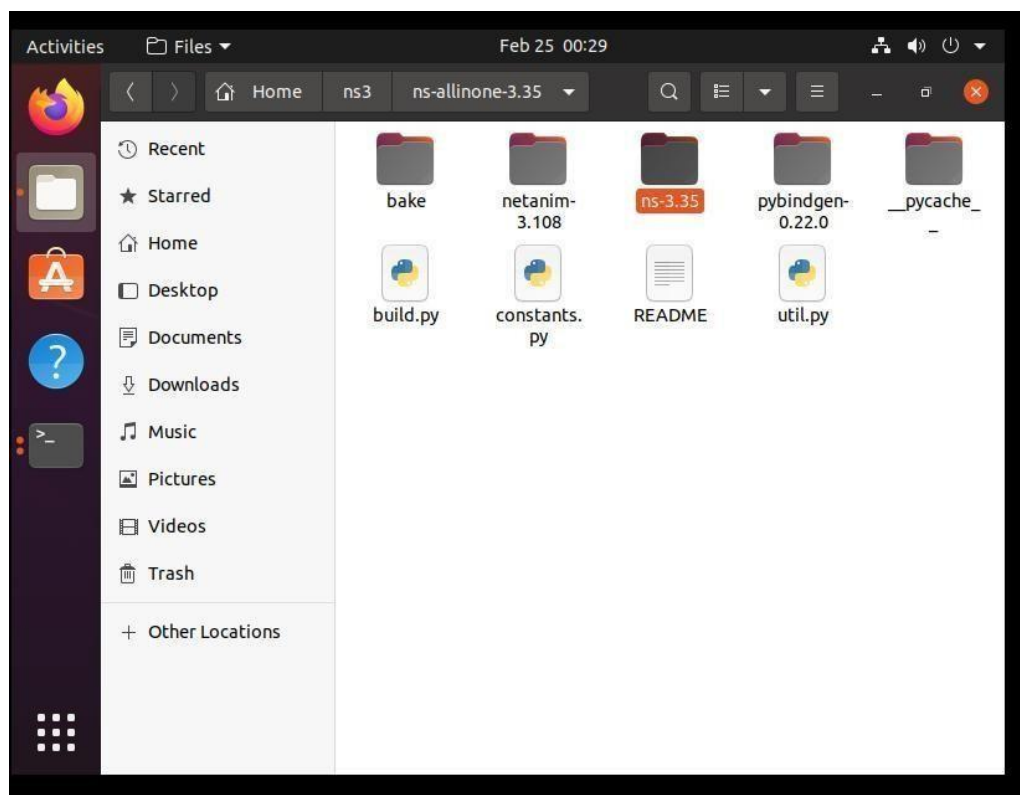
Program to simulate Point to Point Topology.

Start Ubuntu virtual machine , Go to home directory and click on ns3.

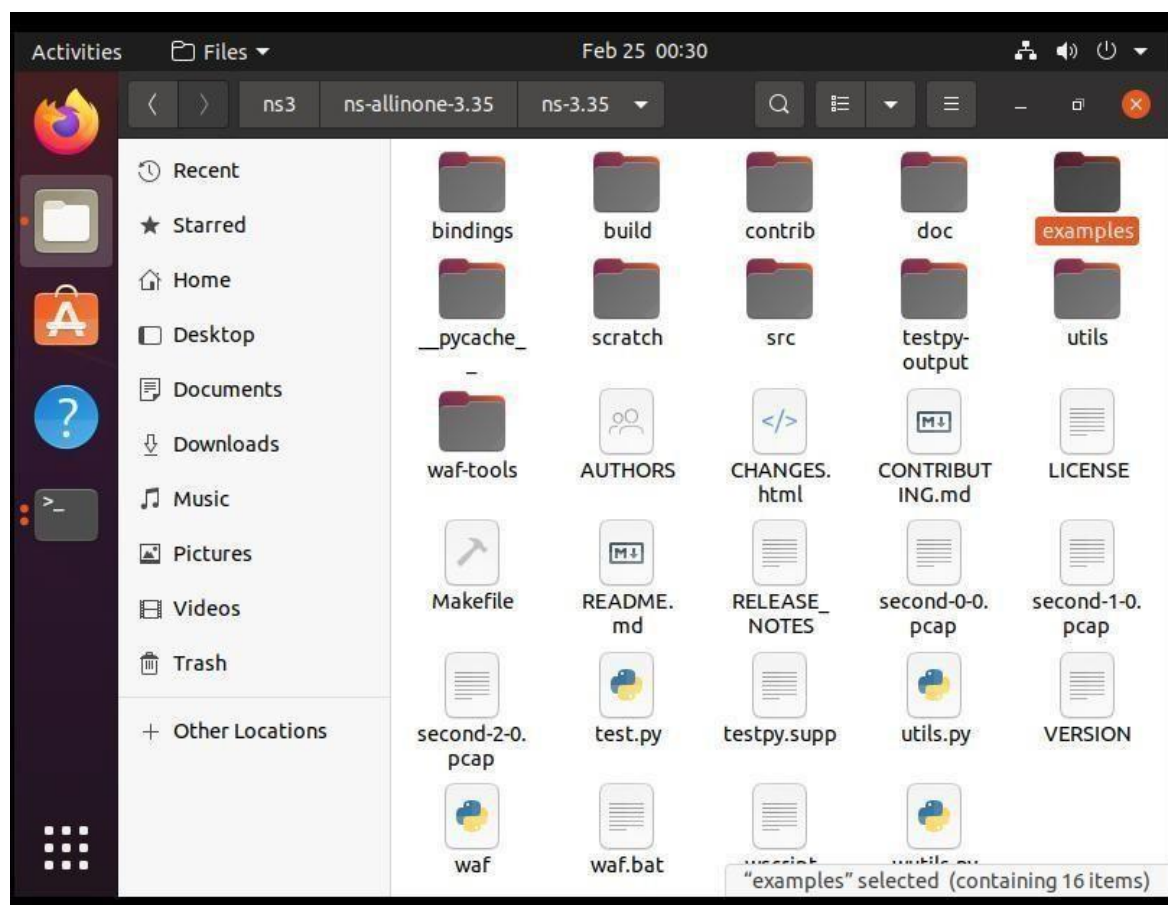


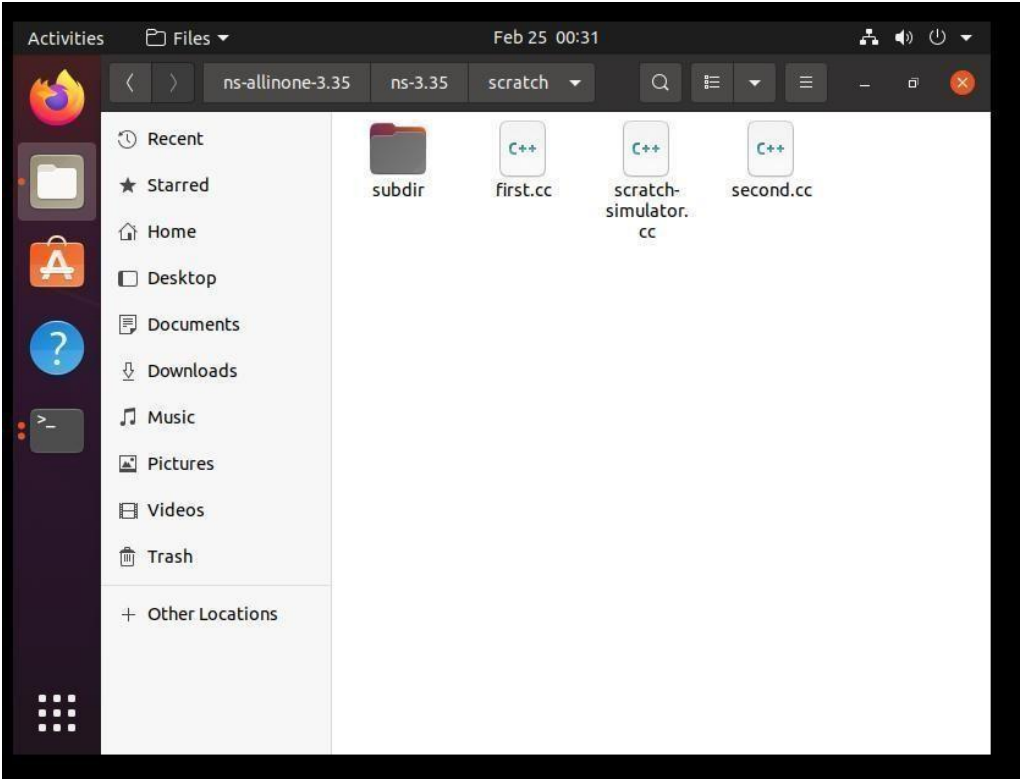
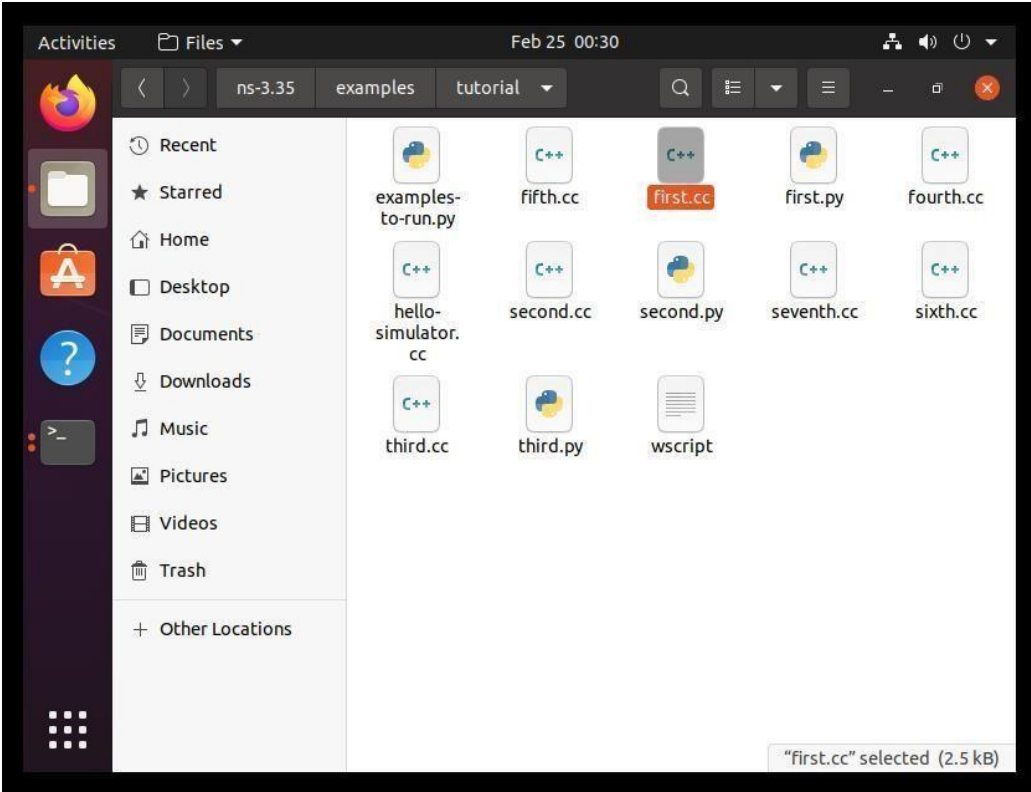
In ns3 folder open nsallinone folder → open ns 3.35 folder.



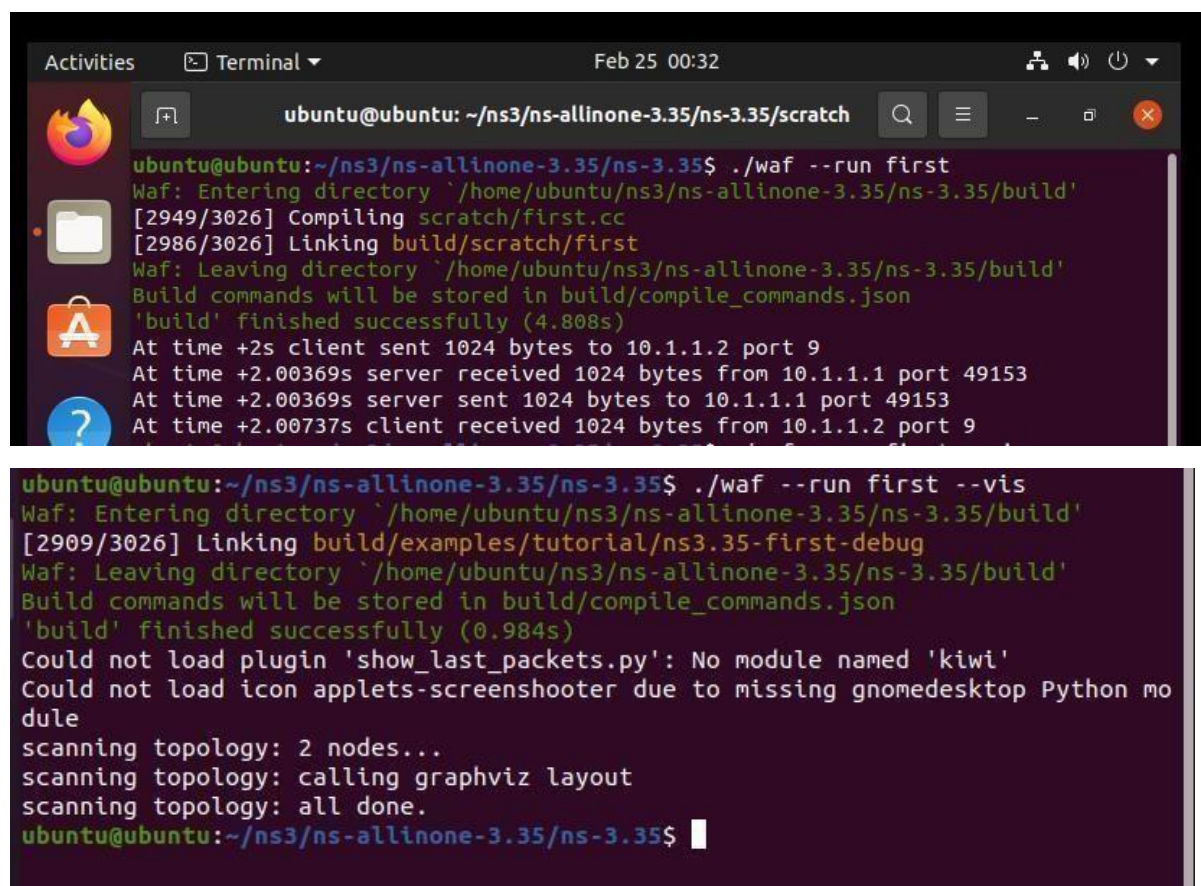


Open examples folder ,go to tutorial and copy the first.cc file into scratch folder of ns3.35.



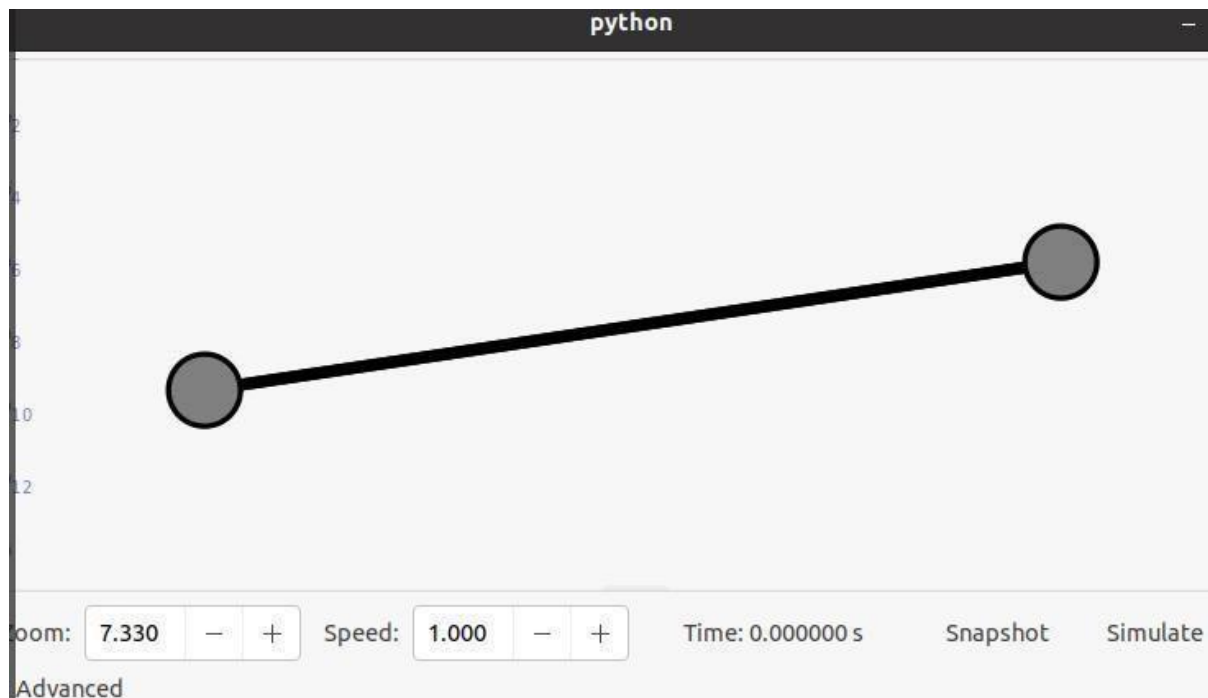


Go to ns 3.35 and then click on open in terminal.



```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35/scratch
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run first
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2949/3026] Compiling scratch/first.cc
[2986/3026] Linking build/scratch/first
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (4.808s)
At time +2s client sent 1024 bytes to 10.1.1.2 port 9
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9

ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run first --vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2909/3026] Linking build/examples/tutorial/ns3.35-first-debug
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.984s)
Could not load plugin 'show_last_packets.py': No module named 'kiwi'
Could not load icon applets-screenshooter due to missing gnomedesktop Python module
scanning topology: 2 nodes...
scanning topology: calling graphviz layout
scanning topology: all done.
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$
```

**CODE:**

```
/* -- Mode:C++; c-file-style:"gnu"; indent-tabs-mode:nil; -- /
/
```

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- */

```
#include "ns3/core-module.h"
#include
"ns3/network-module.h"
```



```
#include "ns3/internet-module.h" #
include
"ns3/point-to-point-module.h" #
include
"ns3/applications-module.h"
//netanimation
#include
"ns3/netanim-module.h"
#include
"ns3/mobility-module.h"
[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]

// Default Network Topology
//
// 10.1.1.0
// n0 n1
// point-to-point
//

using namespace ns3;
NS_LOG_COMPONENT_DEFINE ("FirstScriptExample");
int
main (int argc, char *argv[])
{
CommandLine cmd ( FILE );
cmd.Parse (argc, argv);

Time::SetResolution (Time::NS);
LogComponentEnable ("UdpEchoClientApplication", LOG_LEVEL_INFO);
LogComponentEnable ("UdpEchoServerApplication", LOG_LEVEL_INFO);

NodeContainer nodes;
nodes.Create (2);

PointToPointHelper pointToPoint;
pointToPoint.SetDeviceAttribute ("DataRate", StringValue
("5Mbps")); pointToPoint.SetChannelAttribute ("Delay",
StringValue ("2ms"));

NetDeviceContainer devices;
devices = pointToPoint.Install (nodes);

InternetStackHelper stack;
stack.Install (nodes);

Ipv4AddressHelper address;
address.SetBase ("10.1.1.0", "255.255.255.0");

Ipv4InterfaceContainer interfaces = address.Assign (devices);

UdpEchoServerHelper echoServer (9);
```

```
ApplicationContainer serverApps = echoServer.Install
(nodes.Get (1)); serverApps.Start (Seconds (1.0));
serverApps.Stop (Seconds (10.0));

UdpEchoClientHelper echoClient (interfaces.GetAddress (1), 9);
echoClient.SetAttribute ("MaxPackets", UIntegerValue (1));
echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));
echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));

ApplicationContainer clientApps = echoClient.Install (nodes.Get
(0)); clientApps.Start (Seconds (2.0));
clientApps.Stop (Seconds (10.0));

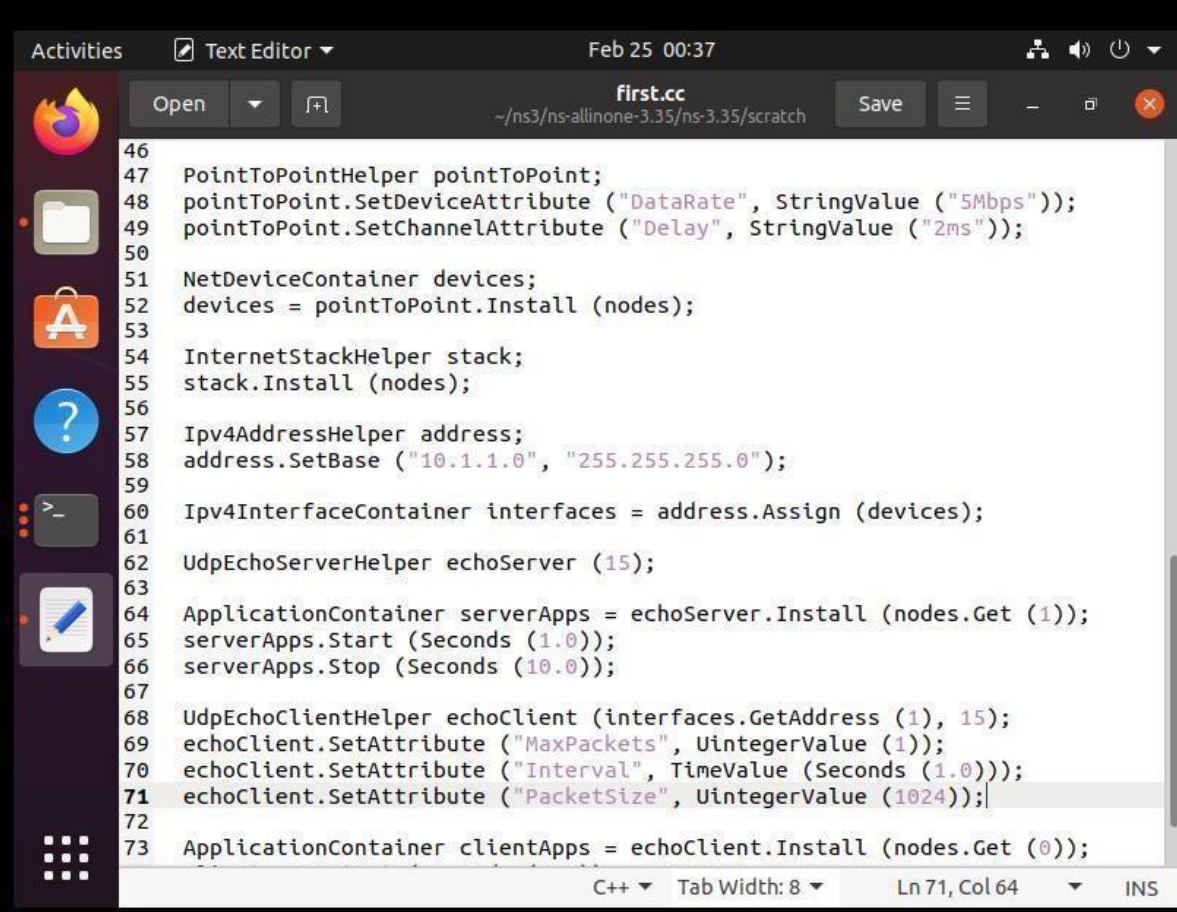
MobilityHelper mobility;
mobility.SetMobilityModel("ns3::ConstantPositionMobilityModel
"); mobility.Install(nodes);
AnimationInterface anim("first.xml");
AnimationInterface::SetConstantPosition(nodes.Get(0),
10,25);
AnimationInterface::SetConstantPosition(nodes.Get(1),
40,25); anim.EnablePacketMetadata(true);
pointToPoint.EnablePcapAll("first");
Simulator::Run ();
Simulator::Destroy ();
return 0;
}
```

[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]

This NS-3 code simulates a simple point-to-point network between two nodes, where one node acts as a UDP echo client and the other as a UDP echo server. The nodes are connected via a link with a data rate of 5 Mbps and 2 ms delay. IP addresses are assigned, and the server listens on port 9 while the client sends one 1024-byte packet after 2 seconds. Mobility models fix the nodes' positions, and the simulation includes NetAnim support for visualization and PCAP tracing for packet capture. The simulation runs for 10 seconds, demonstrating basic end-to-end communication, animation, and tracing.

[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]

To edit the given .cc file open the terminal in scratch folder and perform the gedit command.



The screenshot shows a text editor window titled 'first.cc' with the following C++ code:

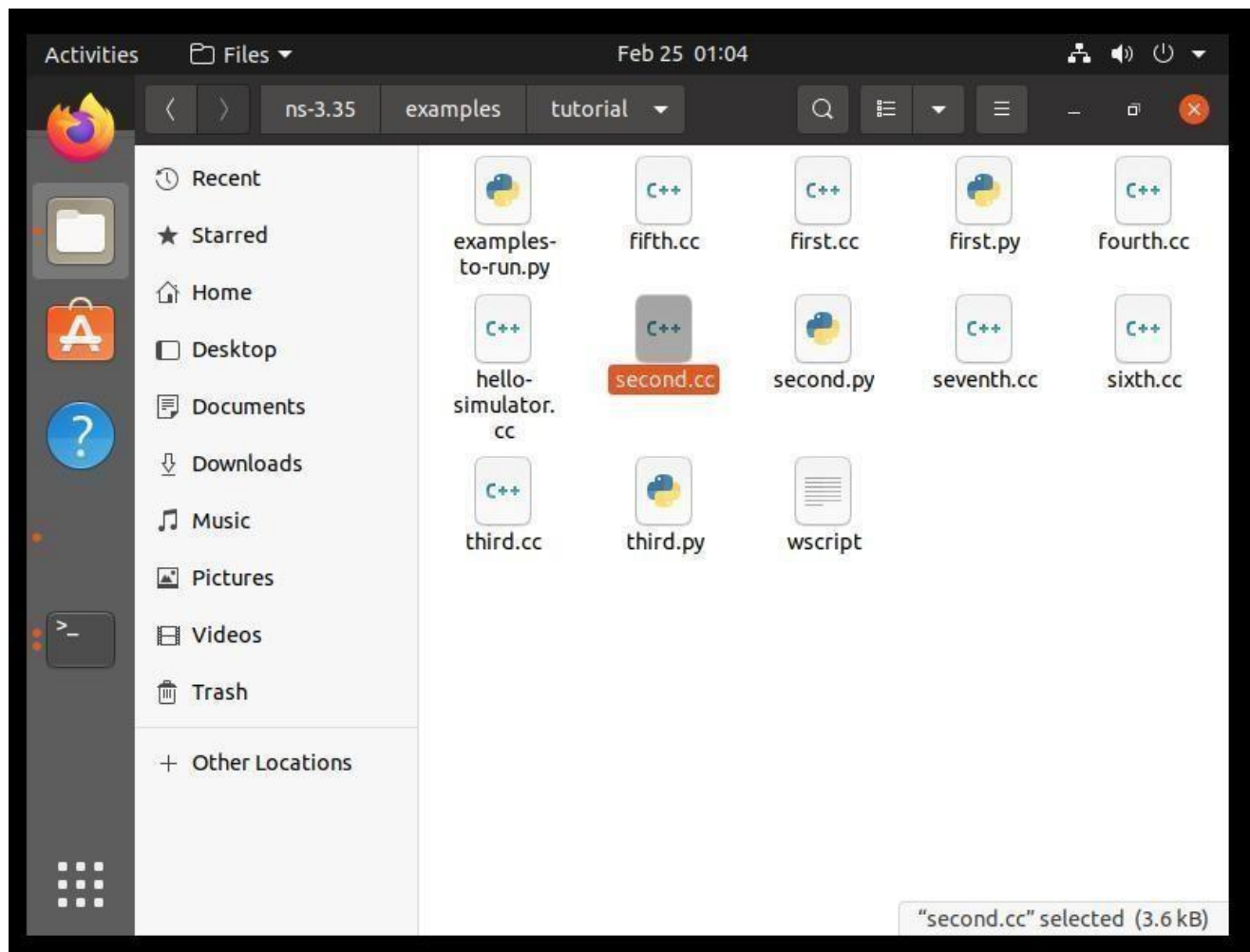
```
46
47 PointToPointHelper pointToPoint;
48 pointToPoint.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
49 pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));
50
51 NetDeviceContainer devices;
52 devices = pointToPoint.Install (nodes);
53
54 InternetStackHelper stack;
55 stack.Install (nodes);
56
57 Ipv4AddressHelper address;
58 address.SetBase ("10.1.1.0", "255.255.255.0");
59
60 Ipv4InterfaceContainer interfaces = address.Assign (devices);
61
62 UdpEchoServerHelper echoServer (15);
63
64 ApplicationContainer serverApps = echoServer.Install (nodes.Get (1));
65 serverApps.Start (Seconds (1.0));
66 serverApps.Stop (Seconds (10.0));
67
68 UdpEchoClientHelper echoClient (interfaces.GetAddress (1), 15);
69 echoClient.SetAttribute ("MaxPackets", UIntegerValue (1));
70 echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));
71 echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));
72
73 ApplicationContainer clientApps = echoClient.Install (nodes.Get (0));
```

The editor interface includes a sidebar with application icons, a top bar with 'Open', 'Save', and window controls, and a status bar at the bottom showing 'C++', 'Tab Width: 8', 'Ln 71, Col 64', and 'INS'.

```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run scratch/first
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2912/3026] Linking build/examples/tutorial/ns3.35-first-debug
[2956/3026] Compiling scratch/first.cc
[2986/3026] Linking build/scratch/first
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (5.326s)
At time +2s client sent 1024 bytes to 10.1.1.2 port 15
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 15
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$
```

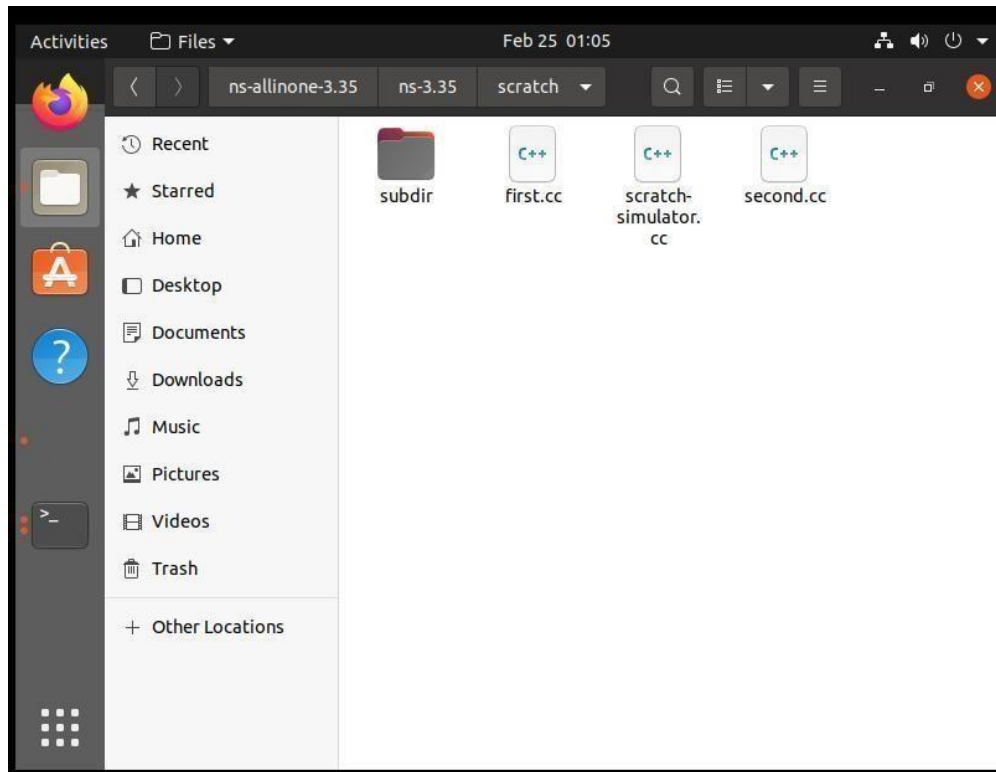
PRACTICAL NO. 4

Program to simulate Bus topology

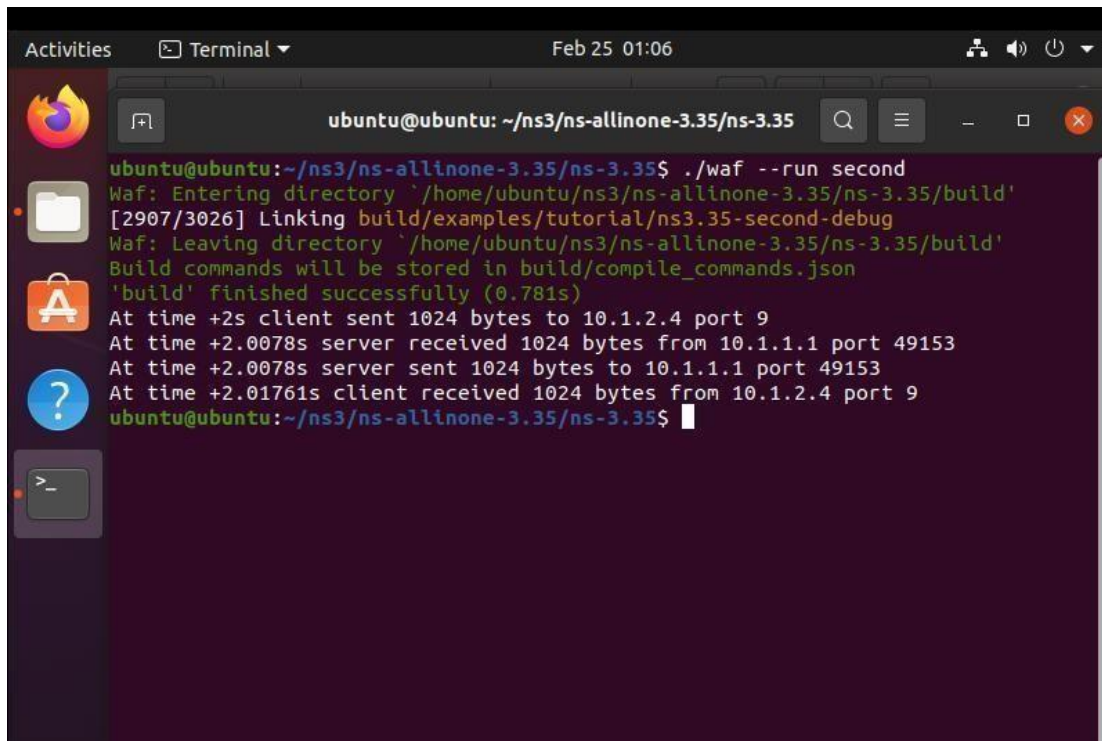


Open the ns3.35 folder and in examples → tutorial → second.cc

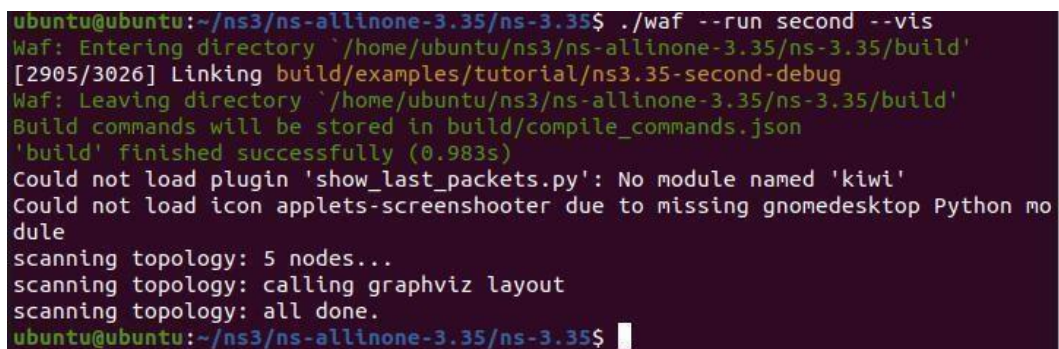
Copy this file and paste it in scratch folder.



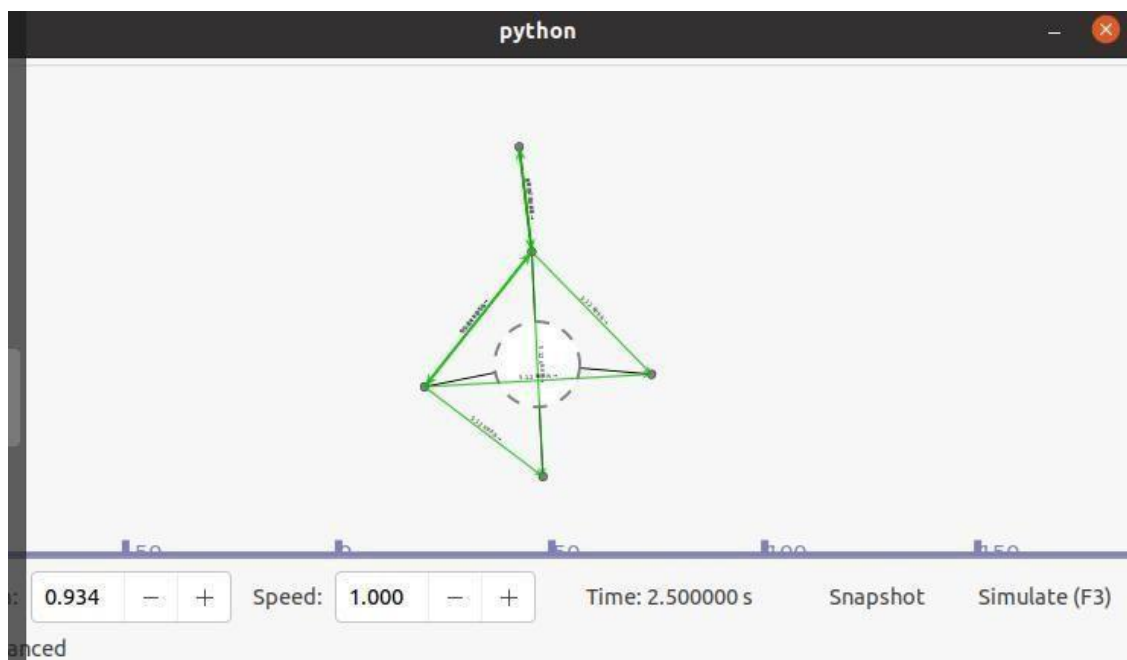
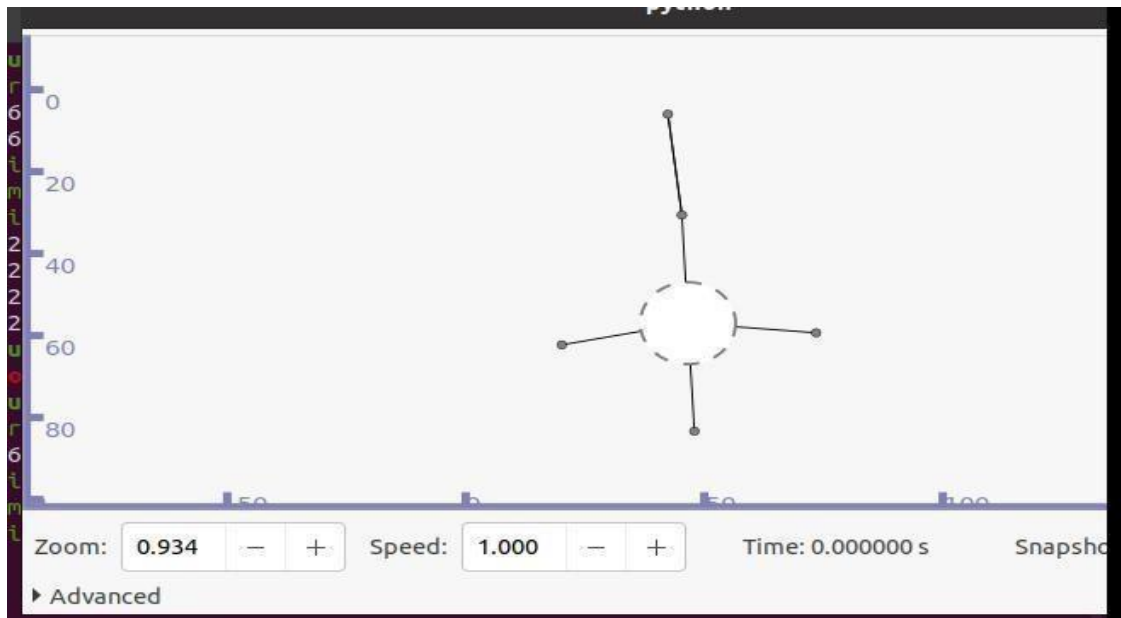
Open the terminal in ns3.35 and perform the commands.



```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run second
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2907/3026] Linking build/examples/tutorial/ns3.35-second-debug
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.781s)
At time +2s client sent 1024 bytes to 10.1.2.4 port 9
At time +2.0078s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.0078s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.01761s client received 1024 bytes from 10.1.2.4 port 9
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$
```



```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run second --vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2905/3026] Linking build/examples/tutorial/ns3.35-second-debug
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.983s)
Could not load plugin 'show_last_packets.py': No module named 'kiwi'
Could not load icon applets-screenshooter due to missing gnomedesktop Python module
scanning topology: 5 nodes...
scanning topology: calling graphviz layout
scanning topology: all done.
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$
```

**CODE:**

```
/* -- Mode:C++; c-file-style:"gnu"; indent-tabs-mode:nil; -- */
```

```
/*
```

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*

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*

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* along with this program; if not, write to the Free Software
* Foundation, Inc., 59 Temple Place, Suite 330, Boston, MA 02111-1307 USA
*/

```
#include "ns3/core-module.h"
```

```
#include
```

```
"ns3/network-module.h"
```

```
#include
```

```
"ns3/csma-module.h"
```

```
#include
```

```
"ns3/internet-module.h"
```

```
#include "ns3/point-to-point-module.h" #
```

```
include "ns3/applications-module.h" #
```

```
include
```

```
"ns3/ipv4-global-routing-helper.h"
```

```
// Default Network Topology
```

```
//
```

```
// 10.1.1.0
```

```
// n0 ----- n1 n2 n3 n4
```

```
// point-to-point | | | |
```

```
// =====
```

```
// LAN 10.1.2.0
```

```
using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("SecondScriptExample");

int
main (int argc, char *argv[])
{
    bool verbose =
    true; uint32_t
    nCsma = 3;

    CommandLine cmd (_FILE_);
    cmd.AddValue ("nCsma", "Number of \"extra\" CSMA nodes/devices", nCsma);
    cmd.AddValue ("verbose", "Tell echo applications to log if true", verbose);

    cmd.Parse (argc,argv);

    if (verbose)
    {
        LogComponentEnable ("UdpEchoClientApplication", LOG_LEVEL_INFO);
        LogComponentEnable ("UdpEchoServerApplication", LOG_LEVEL_INFO);
    }

    nCsma = nCsma == 0 ? 1 : nCsma;

    NodeContainer p2pNodes;
    p2pNodes.Create (2);

    NodeContainer csmaNodes;
    csmaNodes.Add (p2pNodes.Get
(1)); csmaNodes.Create
```

```
(nCsmas);
```

```
PointToPointHelper pointToPoint;
```

```
pointToPoint.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));  
pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));
```

```
NetDeviceContainer p2pDevices;
```

```
p2pDevices = pointToPoint.Install (p2pNodes);
```

```
CsmaHelper csma;
```

```
csma.SetChannelAttribute ("DataRate", StringValue ("100Mbps"));
```

```
csma.SetChannelAttribute ("Delay", TimeValue (NanoSeconds (6560))));
```

```
NetDeviceContainer csmaDevices;
```

```
csmaDevices = csma.Install
```

```
(csmaNodes);
```

```
InternetStackHelper stack;
```

```
stack.Install (p2pNodes.Get
```

```
(0)); stack.Install
```

```
(csmaNodes);
```

```
Ipv4AddressHelper address;
```

```
address.SetBase ("10.1.1.0", "255.255.255.0");
```

```
Ipv4InterfaceContainer p2pInterfaces;
```

```
p2pInterfaces = address.Assign
```

```
(p2pDevices);
```

```
address.SetBase ("10.1.2.0", "255.255.255.0");
```

```
Ipv4InterfaceContainer csmaInterfaces;
```

```
csmaInterfaces = address.Assign
```

```
(csmaDevices);
```



```
UdpEchoServerHelper echoServer (9);

ApplicationContainer serverApps = echoServer.Install (csmaNodes.Get (nCsma));
serverApps.Start (Seconds (1.0));

serverApps.Stop (Seconds (10.0));

UdpEchoClientHelper echoClient (csmaInterfaces.GetAddress (nCsma),
9); echoClient.SetAttribute ("MaxPackets", UIntegerValue (1));
echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));
echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));

ApplicationContainer clientApps = echoClient.Install (p2pNodes.Get (0));
clientApps.Start (Seconds (2.0));
clientApps.Stop (Seconds (10.0));

Ipv4GlobalRoutingHelper::PopulateRoutingTables ();

pointToPoint.EnablePcapAll ("second");
csma.EnablePcap ("second", csmaDevices.Get (1),
true);

Simulator::Run ();
Simulator::Destroy ();
return 0;
}
```

A P2P link connects two nodes: n0 and n1. Node n1 then connects to a group of CSMA nodes (n2, n3, n4, etc.) via a shared CSMA channel, which simulates a bus-like structure where all nodes are connected to the same medium. The CSMA segment acts as the shared backbone, similar to a classic bus topology. The server runs on one of the CSMA nodes, and a client runs on the P2P node n0, sending a UDP echo request to the server. This setup demonstrates communication between

nodes across a mixed topology with bus-like characteristics.

To make changes in .cc file open the file using gedit command.

PRACTICAL NO. 5

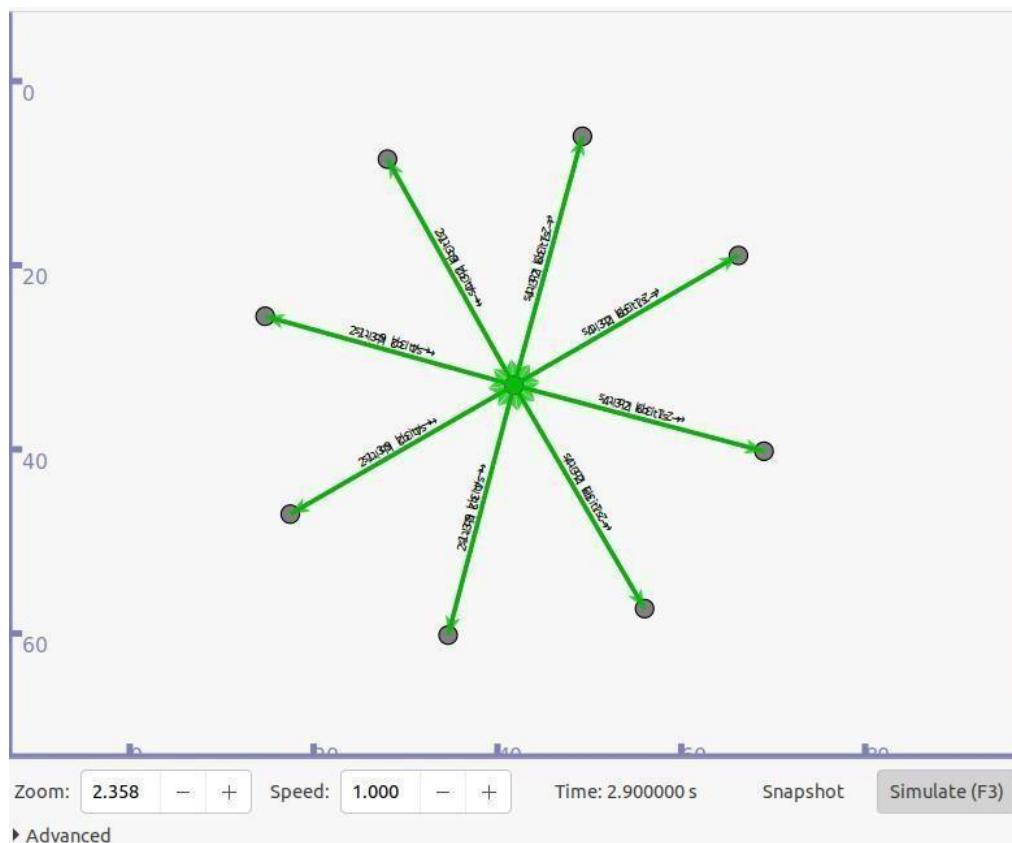
Program to simulate Star topology

Open the ns3.35 folder , go to examples and open tcp folder.

You will find the star.cc there ..Copy the star.cc file to scratch

folder Open the terminal in ns3.35 and compile the star.cc file

```
ubuntu@ubuntu-virtual-machine:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run star
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2895/3028] Linking build/examples/tcp/ns3.35-star-debug
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (1.687s)
ubuntu@ubuntu-virtual-machine:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run star -vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2900/3028] Linking build/examples/tcp/ns3.35-star-debug
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (1.539s)
Could not load plugin 'show_last_packets.py': No module named 'kiwi'
Could not load icon applets-screenshooter due to missing gnomedesktop Python module
scanning topology: 9 nodes...
scanning topology: calling graphviz layout
scanning topology: all done.
```

**CODE:**

```
#include "ns3/core-module.h"
#include
"ns3/network-module.h"
#include
"ns3/netanim-module.h"
#include
"ns3/internet-module.h"
#include "ns3/point-to-point-module.h" #
include "ns3/applications-module.h" #
include
"ns3/point-to-point-layout-module.h"

// Network topology (default)
//
//
```

```
//
n2 n3 n4
\
.
// \/.
// n1--- n0---n5 .
// /\ .
// /\ .
//
//
n8 n7 n6 .
using namespace ns3;
NS_LOG_COMPONENT_DEFINE ("Star");
int
main (int argc, char *argv[])
{

//
// Set up some default values for the simulation.
//
Config::SetDefault ("ns3::OnOffApplication::PacketSize", UIntegerValue (137));

// ??? try and stick 15kb/s into the data rate
Config::SetDefault ("ns3::OnOffApplication::DataRate", StringValue ("14kb/s"));

//
// Default number of nodes in the star. Overridable by command line argument.
//
uint32_t nSpokes = 8;

CommandLine cmd;
```

```
cmd.AddValue ("nSpokes", "Number of nodes to place in the star",  
nSpokes); cmd.Parse (argc, argv);
```

```
NS_LOG_INFO ("Build star  
topology."); PointToPointHelper  
pointToPoint;
```

```
pointToPoint.SetDeviceAttribute ("DataRate", StringValue  
("5Mbps")); pointToPoint.SetChannelAttribute ("Delay",  
StringValue ("2ms")); PointToPointStarHelper star (nSpokes,  
pointToPoint);
```

```
NS_LOG_INFO ("Install internet stack on all nodes.");  
InternetStackHelper internet;  
star.InstallStack (internet);
```

```
NS_LOG_INFO ("Assign IP Addresses.");  
star.AssignIpv4Addresses (Ipv4AddressHelper ("10.1.1.0", "255.255.255.0"));
```

```
NS_LOG_INFO ("Create applications.");  
//  
// Create a packet sink on the star "hub" to receive packets.  
//  
uint16_t port = 50000;  
Address hubLocalAddress (InetSocketAddress (Ipv4Address::GetAny (),  
port)); PacketSinkHelper packetSinkHelper ("ns3::TcpSocketFactory",  
hubLocalAddress); ApplicationContainer hubApp = packetSinkHelper.Install  
(star.GetHub ()); hubApp.Start (Seconds (1.0));  
hubApp.Stop (Seconds (10.0));
```



```
//  
// Create OnOff applications to send TCP to the hub, one on each spoke node.  
//  
OnOffHelper onOffHelper ("ns3::TcpSocketFactory", Address ());  
  
onOffHelper.SetAttribute ("OnTime", StringValue  
("ns3::ConstantRandomVariable[Constant=1]"));  
onOffHelper.SetAttribute ("OffTime", StringValue  
("ns3::ConstantRandomVariable[Constant=0]"));  
  
ApplicationContainer spokeApps;  
  
for (uint32_t i = 0; i < star.SpokeCount (); ++i)  
{  
    AddressValue remoteAddress (InetSocketAddress  
(star.GetHubIpv4Address (i), port));  
    onOffHelper.SetAttribute ("Remote", remoteAddress);  
    spokeApps.Add (onOffHelper.Install (star.GetSpokeNode  
(i)));  
}  
spokeApps.Start (Seconds (1.0));  
spokeApps.Stop (Seconds (10.0));  
  
NS_LOG_INFO ("Enable static global routing.");  
//  
// Turn on global static routing so we can actually be routed across the star.  
//  
Ipv4GlobalRoutingHelper::PopulateRoutingTables ();  
  
NS_LOG_INFO ("Enable pcap tracing.");  
//
```

```
// Do pcap tracing on all point-to-point devices on all nodes.
```

```
//
```

```
pointToPoint.EnablePcapAll ("star");
```

```
NS_LOG_INFO ("Run Simulation.");
```

```
Simulator::Run ();
```

```
Simulator::Destroy ();
```

```
NS_LOG_INFO ("Done.");
```

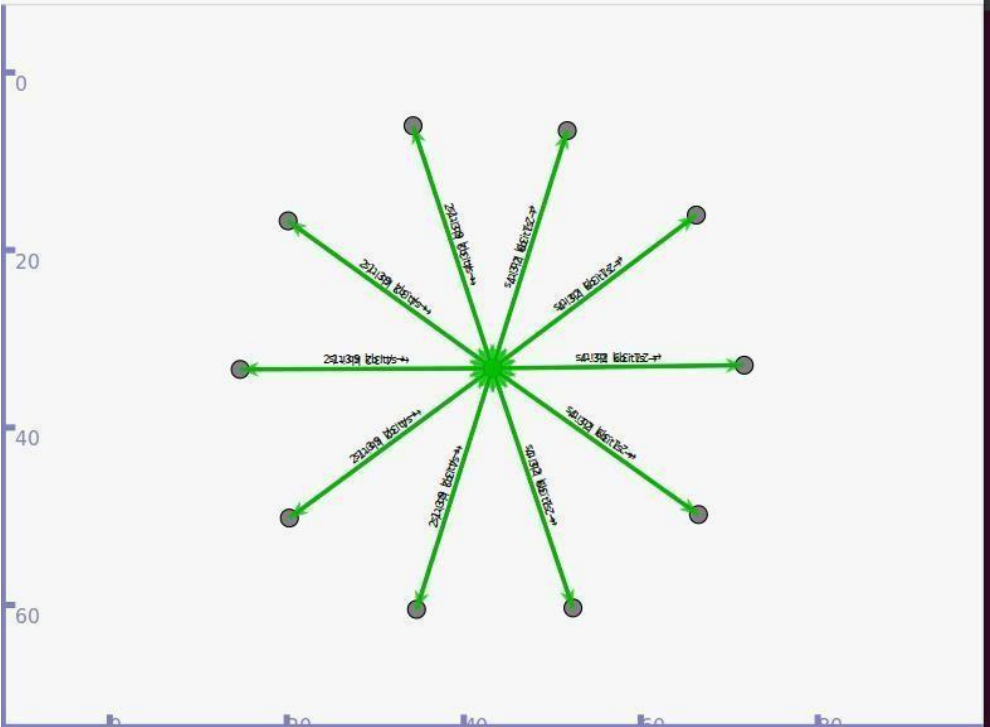
```
return 0;
```

```
}
```

This NS-3 simulation code creates a star network topology with one central hub node connected to multiple spoke nodes (default 8). Each spoke node runs an OnOff TCP application sending traffic to a packet sink application on the hub node. The point-to-point links have 5 Mbps bandwidth and 2 ms delay. The simulation runs from 1 to 10 seconds, with global static routing enabled to route packets properly. It also enables pcap tracing to capture packet data for analysis. Overall, it demonstrates basic TCP communication in a star topology with traffic flowing from multiple spoke nodes to a single central hub.

To make changes in code use gedit command to open the cc file and make changes in it.

```
ubuntu@ubuntu-virtual-machine:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run scratch/star
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2894/3028] Linking build/examples/tcp/ns3.35-star-debug
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (1.447s)
ubuntu@ubuntu-virtual-machine:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run scratch/star --vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.849s)
Could not load plugin 'show_last_packets.py': No module named 'kiwi'
Could not load icon applets-screenshooter due to missing gnomedesktop Python module
scanning topology: 11 nodes...
scanning topology: calling graphviz layout
scanning topology: all done.
ubuntu@ubuntu-virtual-machine:~/ns3/ns-allinone-3.35/ns-3.35$
```



PRACTICAL NO. 6

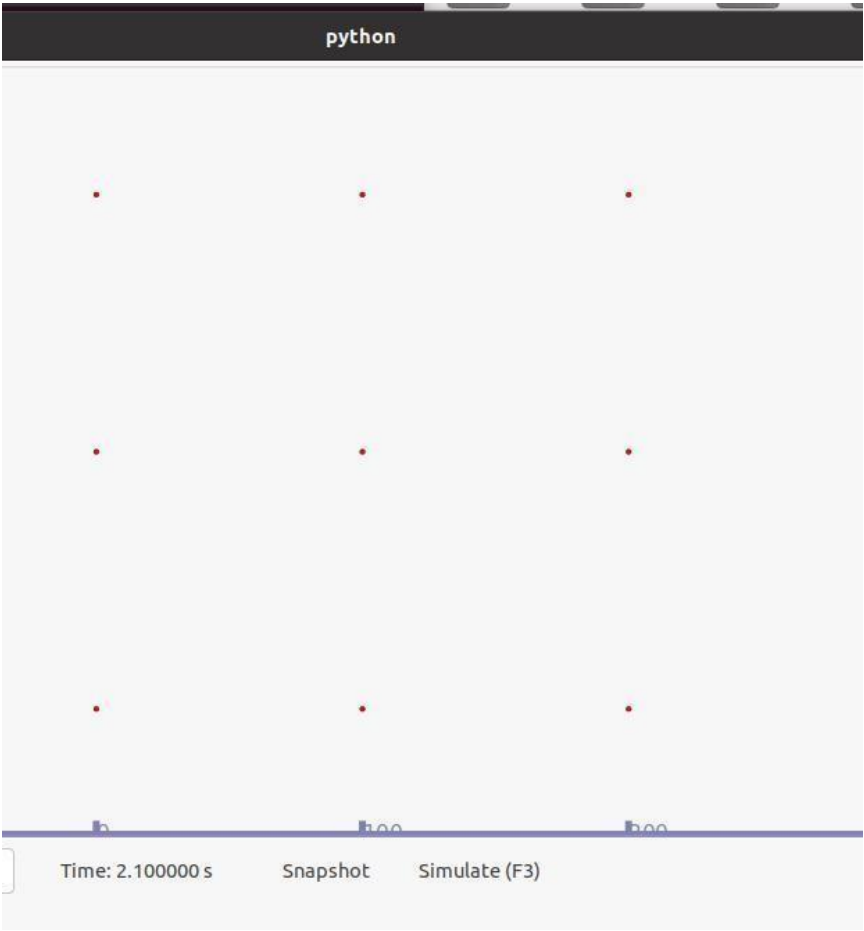
Program to simulate mesh topology.

Open ns3.35→src→mesh→examples→mesh.cc Copy the mesh.cc file to scratch folder

Open terminal in ns3.35 and compile the mesh.cc file

```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run mesh
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2081/2153] Compiling scratch/mesh.cc
[2112/2153] Linking build/scratch/scratch-simulator
[2113/2153] Linking build/scratch/mesh
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (4.885s)
Printing mesh point device #0 diagnostics to mp-report-0.xml
Printing mesh point device #1 diagnostics to mp-report-1.xml
Printing mesh point device #2 diagnostics to mp-report-2.xml
Printing mesh point device #3 diagnostics to mp-report-3.xml
Printing mesh point device #4 diagnostics to mp-report-4.xml
Printing mesh point device #5 diagnostics to mp-report-5.xml
Printing mesh point device #6 diagnostics to mp-report-6.xml
Printing mesh point device #7 diagnostics to mp-report-7.xml
Printing mesh point device #8 diagnostics to mp-report-8.xml
```

```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run mesh --vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.449s)
could not load plugin 'show_last_packets.py': No module named 'kiwi'
could not load icon applets-screenshooter due to missing gnomedesktop Python mod
```



PRACTICAL NO. 7

Program to simulate Hybrid Topology

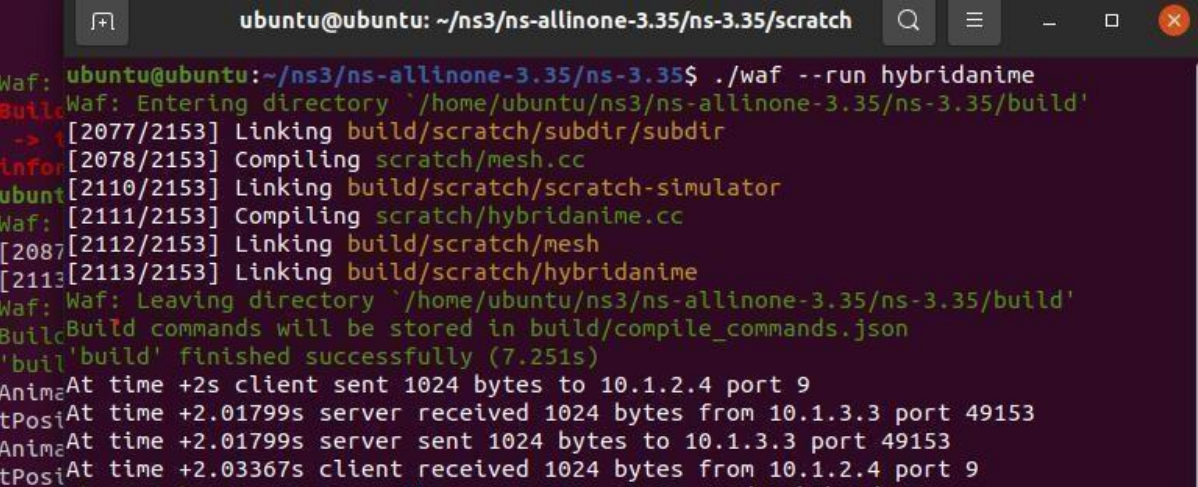
Install the necessary libraries for netanim

```
ubuntu@ubuntu:~$ sudo apt update
[sudo] password for ubuntu:
Hit:1 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  ipython3 python2.7-minimal libcgraph6 python2.7 python3-ipython
  liblab-gamut1 libcdt5 libpathplan4 libgvpr2 libpython2.7-minimal libgvc6
  libpython2.7-stdlib traceroute graphviz
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  ubuntu-pro-client ubuntu-pro-client-l10n
The following packages will be upgraded:
  dns-root-data python3-update-manager snapd ubuntu-advantage-tools
```

```
ubuntu@ubuntu:~$ sudo apt install qt5-default mercurial
Reading package lists... Done
Building dependency tree
Reading state information... Done
mercurial is already the newest version (5.3.1-1ubuntu1).
qt5-default is already the newest version (5.12.8+dfsg-0ubuntu2.1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
ubuntu@ubuntu:~$ sudo apt install gir1.2-gooanvas-2.0 python3-gi python3-gi-cairo python3-pygraphviz gir1.2-gtk-3.0 ipython3
Reading package lists... Done
Building dependency tree
Reading state information... Done
python3-gi is already the newest version (3.36.0-1).
python3-gi-cairo is already the newest version (3.36.0-1).
gir1.2-gooanvas-2.0 is already the newest version (2.0.4-1).
ipython3 is already the newest version (7.13.0-1).
python3-pygraphviz is already the newest version (1.5-4build1).
gir1.2-gtk-3.0 is already the newest version (3.24.20-0ubuntu1.2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Now from examples → tutorial copy paste the third.cc file to scratch Rename the file name to hybridanim and compile the file.



```

ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35/scratch
Waf: ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run hybridanime
Build Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
-> [2077/2153] Linking build/scratch/subdir/subdir
[2078/2153] Compiling scratch/mesh.cc
Infor [2110/2153] Linking build/scratch/scratch-simulator
Waf: [2111/2153] Compiling scratch/hybridanime.cc
[2087 [2112/2153] Linking build/scratch/mesh
[2113 [2113/2153] Linking build/scratch/hybridanime
Waf: Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build Build commands will be stored in build/compile_commands.json
'buil 'build' finished successfully (7.251s)
Anime At time +2s client sent 1024 bytes to 10.1.2.4 port 9
tPosi At time +2.01799s server received 1024 bytes from 10.1.3.3 port 49153
Anime At time +2.01799s server sent 1024 bytes to 10.1.3.3 port 49153
tPosi At time +2.03367s client received 1024 bytes from 10.1.2.4 port 9

```

CODE:

```
/* -*- Mode:C++; c-file-style:"gnu"; indent-tabs-mode:nil; -*- */
```

```
/*
```

```
* This program is free software; you can redistribute it and/or modify
```

```
* it under the terms of the GNU General Public License version 2 as
```

```
* published by the Free Software Foundation;
```

```
*
```

```
* This program is distributed in the hope that it will be useful,
```

```
* but WITHOUT ANY WARRANTY; without even the implied warranty of
```

```
* MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
```

```
* GNU General Public License for more details.
```

```
*
```

```
* You should have received a copy of the GNU General Public License
```

```
* along with this program; if not, write to the Free Software
```

```
* Foundation, Inc., 59 Temple Place, Suite 330, Boston, MA 02111-1307 USA
```

```
*/
```

```
#include "ns3/core-module.h"
```

```
#include
```

```
"ns3/network-module.h"
```

```
#include "ns3/internet-module.h" #
include
"ns3/point-to-point-module.h" #
include
"ns3/applications-module.h"
//netanimation
[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]
#include
"ns3/netanim-module.h"
#include
"ns3/mobility-module.h"

// Default Network Topology
//
// 10.1.1.0
// n0 n1
// point-to-point
//
using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("FirstScriptExample");

int
main (int argc, char *argv[])
{
CommandLine cmd ( FILE );
cmd.Parse (argc, argv);

Time::SetResolution (Time::NS);
LogComponentEnable ("UdpEchoClientApplication", LOG_LEVEL_INFO);
LogComponentEnable ("UdpEchoServerApplication", LOG_LEVEL_INFO);
```

```
NodeContainer nodes;  
nodes.Create (2);
```

```
PointToPointHelper pointToPoint;  
pointToPoint.SetDeviceAttribute ("DataRate", StringValue  
("5Mbps")); pointToPoint.SetChannelAttribute ("Delay",  
StringValue ("2ms")); NetDeviceContainer devices;  
devices = pointToPoint.Install (nodes);
```

```
InternetStackHelper stack;  
stack.Install (nodes);
```

```
Ipv4AddressHelper address;  
address.SetBase ("10.1.1.0", "255.255.255.0");
```

```
Ipv4InterfaceContainer interfaces = address.Assign (devices);
```

```
UdpEchoServerHelper echoServer (9);
```

```
ApplicationContainer serverApps = echoServer.Install  
(nodes.Get (1)); serverApps.Start (Seconds (1.0));  
serverApps.Stop (Seconds (10.0));
```

```
UdpEchoClientHelper echoClient (interfaces.GetAddress (1), 9);  
echoClient.SetAttribute ("MaxPackets", UIntegerValue (1));  
echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));  
echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));
```

```
ApplicationContainer clientApps = echoClient.Install (nodes.Get  
(0)); clientApps.Start (Seconds (2.0));  
clientApps.Stop (Seconds (10.0));
```

```
MobilityHelper mobility;

mobility.SetMobilityModel("ns3::ConstantPositionMobilityModel
"); mobility.Install(nodes);

AnimationInterface anim("hybridanim.xml");
[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]
anim.SetConstantPosition(p2pNodes.Get(0),10.0,10.0);
anim.SetConstantPosition(p2pNodes.Get(0),20.0,20.0); Simulator::Run
();

Simulator::Destroy ();

return 0;

}
```

Now to make the changes in .cc file open it using gedit and add this lines to create an xml file and save it.

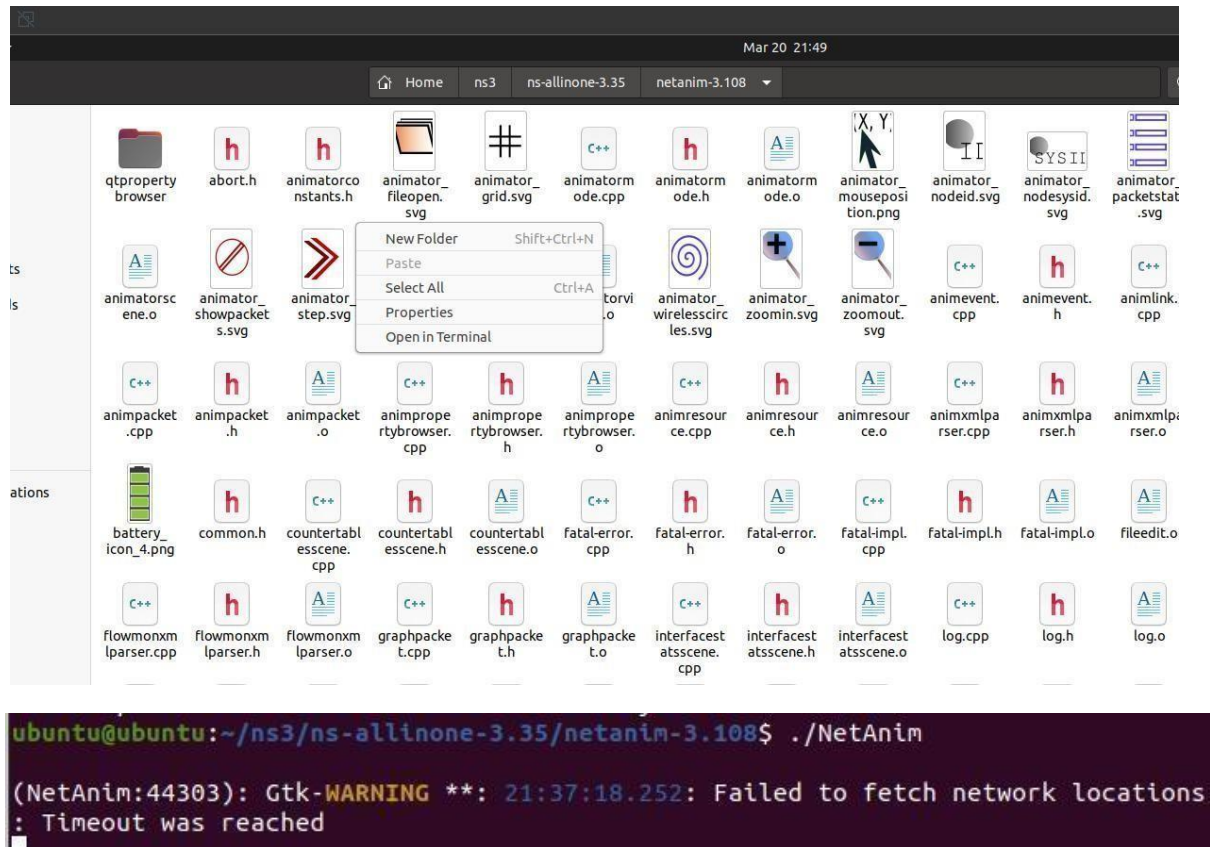
```
    phy.EnablePcap ("third", apDevices.Get (0));
    csma.EnablePcap ("third", csmaDevices.Get (0), true);
}
AnimationInterface anim("hybridanim.xml");
anim.SetConstantPosition(p2pNodes.Get(0),10.0,10.0);
anim.SetConstantPosition(p2pNodes.Get(0),20.0,20.0);
    Simulator::Run ();
    Simulator::Destroy ();
    return 0;
}
```

Compile the saved file.

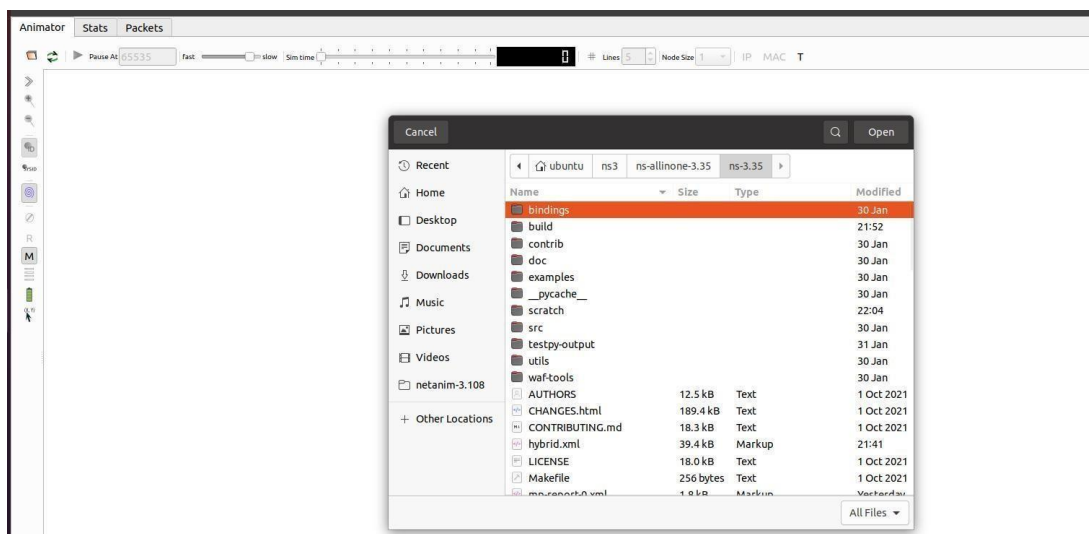
```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run scratch/hybridanim
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2087/2153] Compiling scratch/hybridanim.cc
[2113/2153] Linking build/scratch/hybridanim
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (4.716s)
AnimationInterface WARNING:Node:1 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:1 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstan
tPosition if it is stationary
```

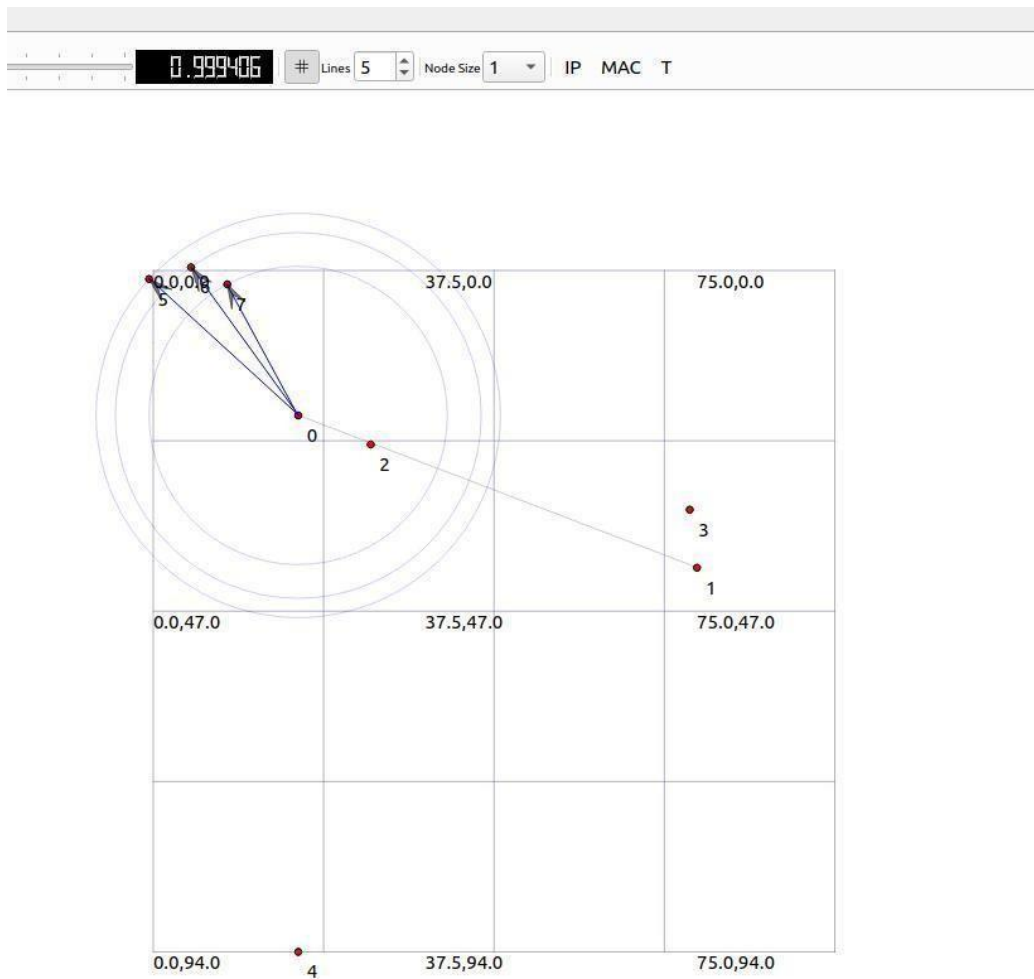
Open Netanim

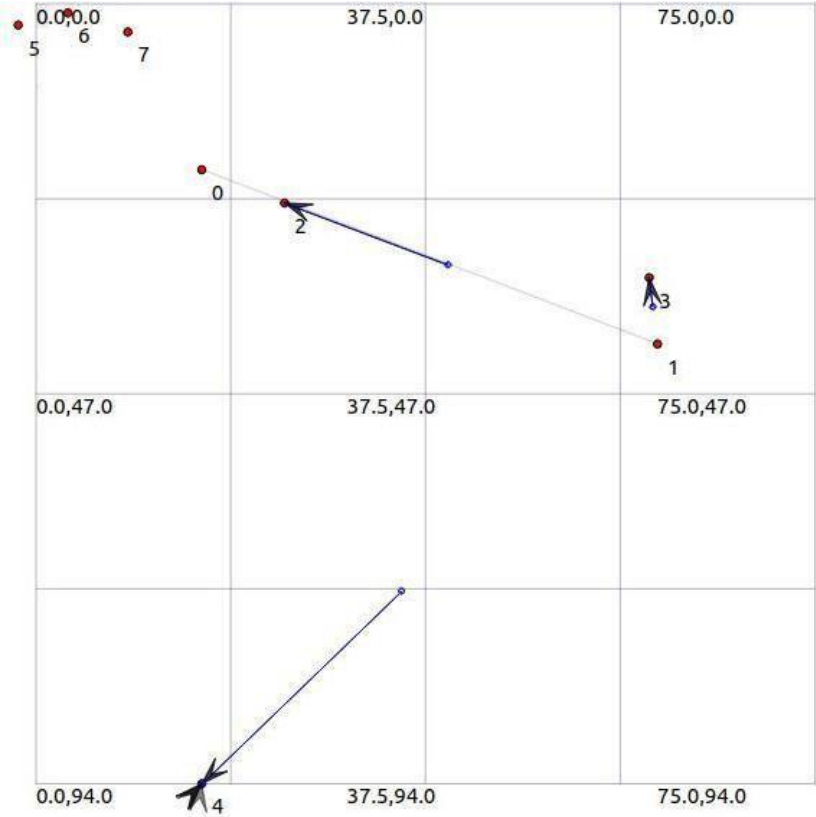




Search for the Xml file created in
Netanim Open the file.







For bus topology

Copy paste third.cc to scratch folder from examples folder and rename it. Compile the file.

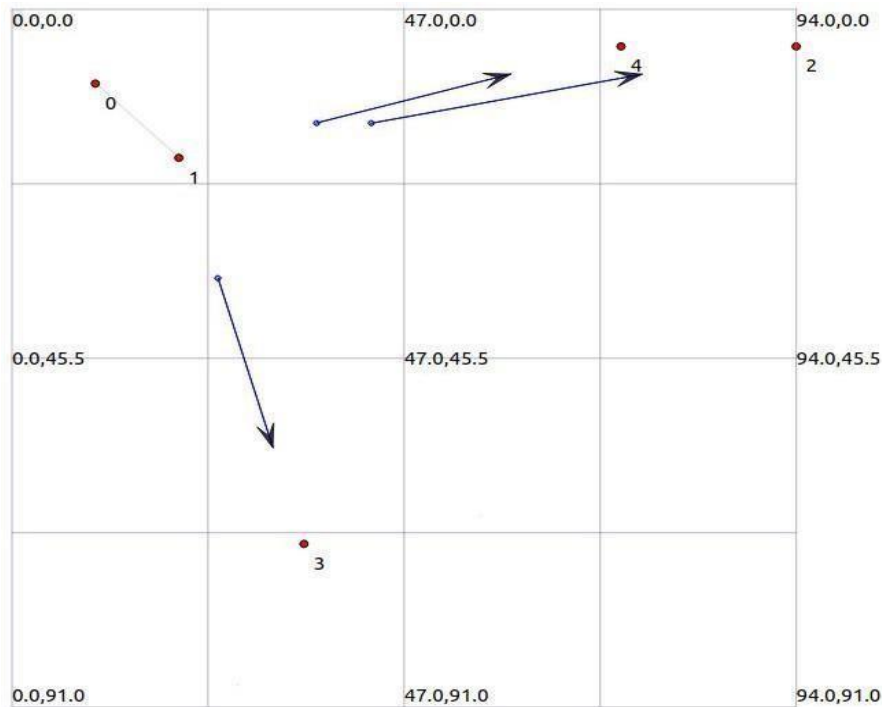
```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35/scratch
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run secondanim
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2079/2155] Linking build/scratch/subdir/subdir
[2080/2155] Compiling scratch/secondanim.cc
[2111/2155] Compiling scratch/hybridanim.cc
[2112/2155] Compiling scratch/scratch-simulator.cc
[2113/2155] Linking build/scratch/secondanim
[2114/2155] Linking build/scratch/scratch-simulator
[2115/2155] Linking build/scratch/hybridanim
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (5.741s)
At time +2s client sent 1024 bytes to 10.1.2.4 port 9
At time +2.0078s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.0078s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.01761s client received 1024 bytes from 10.1.2.4 port 9
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ cd scratch
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35/scratch$ gedit secondanim.cc
```

Make changes in the code and Also write the command to create xml file. Compile the changed file

```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run scratch/secondanim
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2081/2155] Compiling scratch/secondanim.cc
[2115/2155] Linking build/scratch/secondanim
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (4.657s)
AnimationInterface WARNING:Node:0 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:1 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:0 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:1 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
```

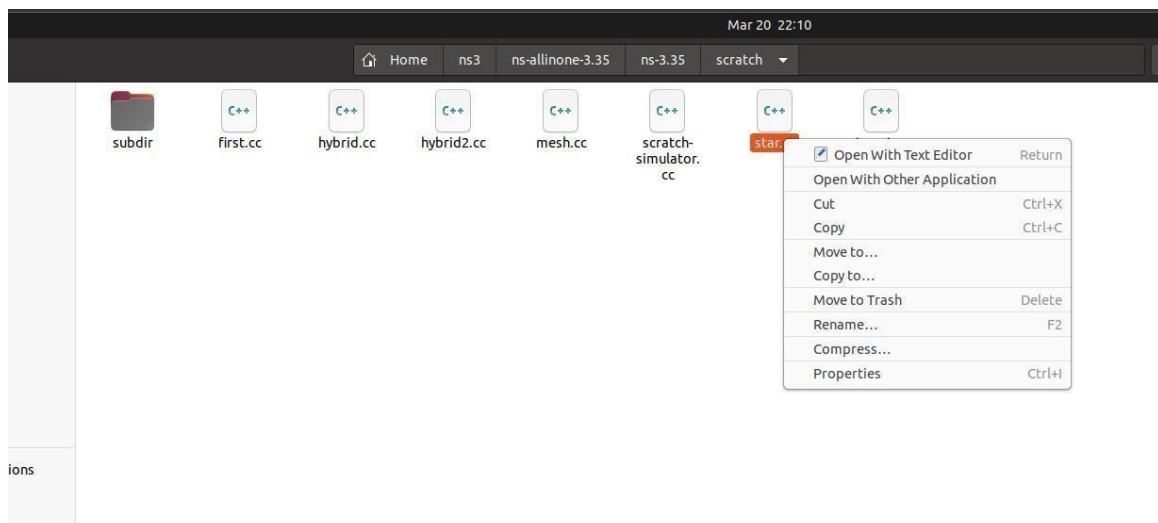
Open Netanim

Search for created xml file and open the xml file.



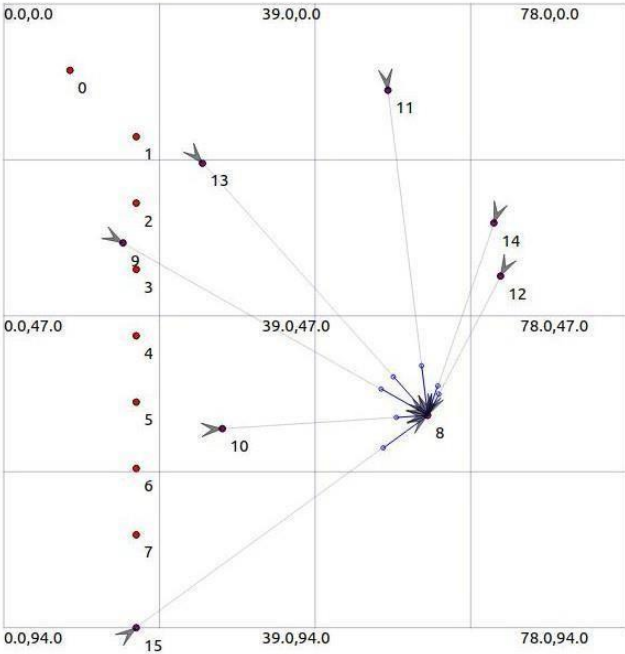
For Netanim Star Topology

Copy paste Star.cc file to scratch folder and rename it.



```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run hybridstar
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2952/3034] Linking build/scratch/scratch-simulator
[2953/3034] Compiling scratch/mesh.cc
[2990/3034] Compiling scratch/hybrid.cc
[2991/3034] Compiling scratch/hybridstar.cc
[2992/3034] Linking build/scratch/mesh
[2993/3034] Linking build/scratch/hybrid
[2994/3034] Linking build/scratch/hybridstar
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (6.709s)
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$
```

```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/netanim-3.108
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/netanim-3.108$ ./NetAnim
```

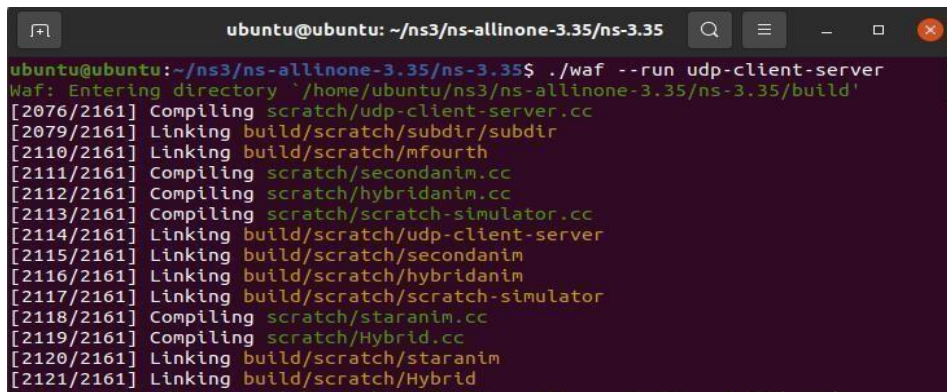



PRACTICAL NO. 8

Program to simulate UDP Client server.

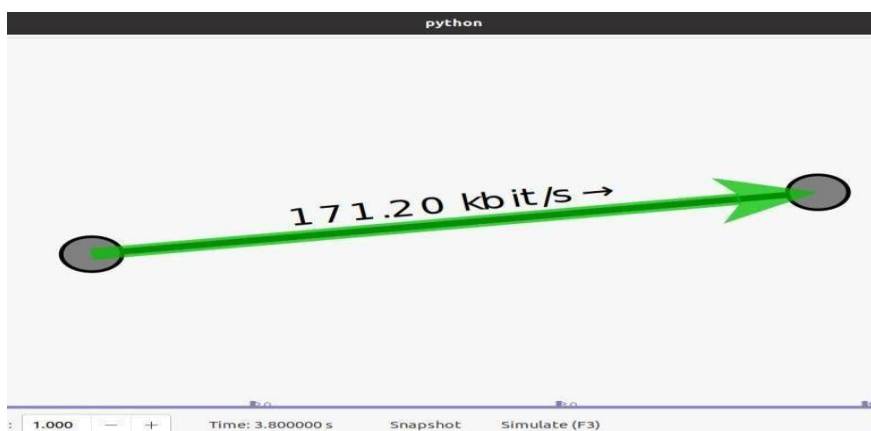
UDP (User Datagram Protocol) is a simple, connectionless transport layer protocol used for fast data transmission. Unlike TCP, it doesn't establish a connection before sending data and doesn't guarantee delivery, order, or error checking. This makes UDP faster and more efficient, ideal for real-time applications like video streaming, online gaming, and VoIP where speed is more important than reliability.

Open the ns3.35 folder and copy paste the udp-client-server file to scratch. Compile the file.



```
ubuntu@ubuntu: ~/ns3/ns-allinone-3.35/ns-3.35
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run udp-client-server
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2076/2161] Compiling scratch/udp-client-server.cc
[2079/2161] Linking build/scratch/subdir/subdir
[2110/2161] Linking build/scratch/mfourth
[2111/2161] Compiling scratch/secondanim.cc
[2112/2161] Compiling scratch/hybridanim.cc
[2113/2161] Compiling scratch/scratch-simulator.cc
[2114/2161] Linking build/scratch/udp-client-server
[2115/2161] Linking build/scratch/secondanim
[2116/2161] Linking build/scratch/hybridanim
[2117/2161] Linking build/scratch/scratch-simulator
[2118/2161] Compiling scratch/staranim.cc
[2119/2161] Compiling scratch/Hybrid.cc
[2120/2161] Linking build/scratch/staranim
[2121/2161] Linking build/scratch/Hybrid
```

```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run udp-client-server --vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.462s)
Could not load plugin 'show_last_packets.py': No module named 'kiwi'
Could not load icon applets-screenshooter due to missing gnomedesktop Python module
scanning topology: 2 nodes...
scanning topology: calling graphviz layout
scanning topology: all done.
TraceDelay TX 1024 bytes to 10.1.1.2 Uid: 0 Time: +2s
TraceDelay: RX 1024 bytes from 10.1.1.1 Sequence Number: 0 Uid: 0 TXtime: +2e+09ns RXtime: +2.01592e+09ns Delay: +1.59188e+07ns
TraceDelay TX 1024 bytes to 10.1.1.2 Uid: 11 Time: +2.05s
TraceDelay: RX 1024 bytes from 10.1.1.1 Sequence Number: 1 Uid: 11 TXtime: +2.05e+09ns RXtime: +2.05371e+09ns Delay: +3.712e+06ns
TraceDelay TX 1024 bytes to 10.1.1.2 Uid: 14 Time: +2.1s
```



PRACTICAL NO. 9

Program to simulate DHCP Server and clients.

DHCP (Dynamic Host Configuration Protocol) is a network protocol used to automatically assign IP addresses and other network settings (like gateway and DNS) to devices on a network. Instead of manually configuring each device, a DHCP server dynamically provides this information, making network management easier and reducing configuration errors. It's commonly used in home routers, offices, and enterprise networks.

CODE:

```
/* -*- Mode:C++; c-file-style:"gnu"; indent-tabs-mode:nil; -*- */  
  
/*  
 * Copyright (c) 2011 UPB  
 * Copyright (c) 2017 NITK Surathkal  
 *  
 * This program is free software; you can redistribute it and/or modify  
 * it under the terms of the GNU General Public License version 2 as  
 * published by the Free Software Foundation;  
 * This program is distributed in the hope that it will be useful,  
 * but WITHOUT ANY WARRANTY; without even the implied warranty of  
 * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the  
 * GNU General Public License for more details.  
 *  
 * You should have received a copy of the GNU General Public License  
 * along with this program; if not, write to the Free Software  
 * Foundation, Inc., 59 Temple Place, Suite 330, Boston, MA 02111-1307 USA  
 *  
 * Author: Radu Lupu <rlupu@elcom.pub.ro>  
 * Ankit Deepak <adadeepak8@gmail.com>  
 * Deepti Rajagopal <deeptir96@gmail.com>  
 *  
 */  
  
/*
```

* Network layout:

*

* R0 is a DHCP server. The DHCP server announced R1 as the default router.

* Nodes N1 will send UDP Echo packets to node A.

*

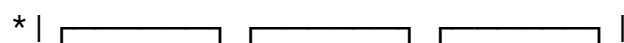
*

* 

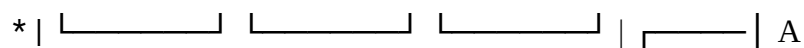
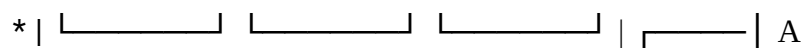
* | DHCP Clients |

* | 172.30.0.14 |

* | DHCP static |

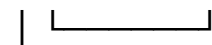
* |  |

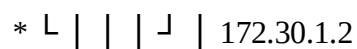
* | | N0 | | N1 | | N2 | |  |

* |  |  | A

|
|
|

| |

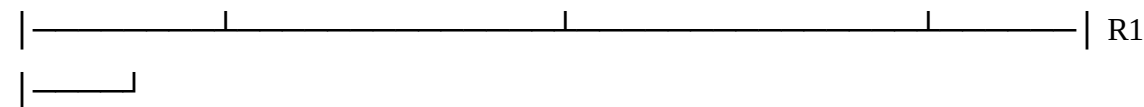
| 

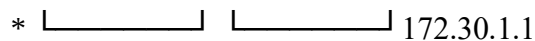
*  | 172.30.1.2

* DHCP Server | | |

*  |

* | R0

|  | R1

*  172.30.1.1

* 172.30.0.12 172.30.0.17

*

* Things to notice:

* 1) The routes in A are manually set to have R1 as the default router,

* just because using a dynamic routing in this example is an overkill.

* 2) R1's address is set statically though the DHCP server helper interface.

- * This is useful to prevent address conflicts with the dynamic pool.
- * Not necessary if the DHCP pool is not conflicting with static addresses.
- * 3) N2 has a dynamically-assigned, static address (i.e., a fixed address)

```
#include "ns3/core-module.h"

#include
"ns3/internet-apps-module.h" #
include "ns3/csma-module.h" #
include "ns3/internet-module.h" #
include
"ns3/point-to-point-module.h" #
include
"ns3/applications-module.h"

//netanim code
#include
"ns3/netanim-module.h"
#include
"ns3/mobility-module.h"

using namespace ns3;
NS_LOG_COMPONENT_DEFINE ("DhcpExample");
int
main (int argc, char *argv[])
{
CommandLine cmd ( FILE );

bool verbose =
false; bool tracing
= false;
```



```
cmd.AddValue ("verbose", "turn on the logs", verbose);
cmd.AddValue ("tracing", "turn on the tracing", tracing);

cmd.Parse (argc, argv);
// GlobalValue::Bind ("ChecksumEnabled", BooleanValue
(true)); if (verbose)
{
LogComponentEnable ("DhcpServer", LOG_LEVEL_ALL);
LogComponentEnable ("DhcpClient", LOG_LEVEL_ALL); LogComponentEnable
("UdpEchoServerApplication", LOG_LEVEL_INFO); LogComponentEnable
("UdpEchoClientApplication", LOG_LEVEL_INFO);
}

Time stopTime = Seconds (20);

NS_LOG_INFO ("Create
nodes."); NodeContainer nodes;
NodeContainer router;
nodes.Create (3);
router.Create (2);

NodeContainer net (nodes, router);

NS_LOG_INFO ("Create channels.");
CsmaHelper csma;

csma.SetChannelAttribute ("DataRate", StringValue
("5Mbps")); csma.SetChannelAttribute ("Delay",
StringValue ("2ms")); csma.SetDeviceAttribute ("Mtu",
UIntegerValue (1500)); NetDeviceContainer devNet =
csma.Install (net);
```

```
NodeContainer p2pNodes;  
p2pNodes.Add (net.Get (4));  
p2pNodes.Create (1);
```

```
PointToPointHelper pointToPoint;  
pointToPoint.SetDeviceAttribute ("DataRate", StringValue  
("5Mbps")); pointToPoint.SetChannelAttribute ("Delay",  
StringValue ("2ms"));
```

```
NetDeviceContainer p2pDevices;  
  
p2pDevices = pointToPoint.Install (p2pNodes);
```

```
InternetStackHelper tcpip;  
tcpip.Install (nodes);  
tcpip.Install (router);  
tcpip.Install (p2pNodes.Get  
(1));
```

```
Ipv4AddressHelper address;  
address.SetBase ("172.30.1.0", "255.255.255.0");  
Ipv4InterfaceContainer p2pInterfaces;  
p2pInterfaces = address.Assign  
(p2pDevices);
```

```
// manually add a routing entry because we don't want to add a dynamic routing
```

```
Ipv4StaticRoutingHelper ipv4RoutingHelper;  
Ptr<Ipv4> ipv4Ptr = p2pNodes.Get (1)->GetObject<Ipv4> ();  
Ptr<Ipv4StaticRouting> staticRoutingA =  
ipv4RoutingHelper.GetStaticRouting (ipv4Ptr);  
staticRoutingA->AddNetworkRouteTo (Ipv4Address ("172.30.0.0"), Ipv4Mask  
("/24"), Ipv4Address ("172.30.1.1"), 1);
```

```
NS_LOG_INFO ("Setup the IP addresses and create DHCP applications.");
DhcpHelper dhcpHelper;

// The router must have a fixed IP.

Ipv4InterfaceContainer fixedNodes = dhcpHelper.InstallFixedAddress (devNet.Get
(4), Ipv4Address ("172.30.0.17"), Ipv4Mask ("/24"));
// Not really necessary, IP forwarding is enabled by default in IPv4.
FixedNodes.Get (0).first->SetAttribute ("IpForward", BooleanValue (true));

// DHCP server
ApplicationContainer dhcpServerApp = dhcpHelper.InstallDhcpServer (devNet.Get
(3), Ipv4Address ("172.30.0.12"),

("/24"),
("172.30.0.15"),
Ipv4Address ("172.30.0.0"), Ipv4Mask

Ipv4Address ("172.30.0.10"), Ipv4Address

Ipv4Address ("172.30.0.17"));

// This is just to show how it can be done.
DynamicCast<DhcpServer> (dhcpServerApp.Get (0))->AddStaticDhcpEntry
(devNet.Get (2)->GetAddress (), Ipv4Address ("172.30.0.14"));

dhcpServerApp.Start (Seconds
(0.0)); dhcpServerApp.Stop
(stopTime);
```

```
// DHCP clients
```

```
NetDeviceContainer dhcpClientNetDevs;  
dhcpClientNetDevs.Add (devNet.Get (0));  
dhcpClientNetDevs.Add (devNet.Get (1));  
dhcpClientNetDevs.Add (devNet.Get (2));
```

```
ApplicationContainer dhcpClients =  
dhcpHelper.InstallDhcpClient (dhcpClientNetDevs);  
dhcpClients.Start (Seconds (1.0));  
dhcpClients.Stop (stopTime);
```

```
UdpEchoServerHelper echoServer (9);
```

```
ApplicationContainer serverApps = echoServer.Install (p2pNodes.Get (1));  
serverApps.Start (Seconds (0.0));  
serverApps.Stop (stopTime);
```

```
UdpEchoClientHelper echoClient (p2pInterfaces.GetAddress (1), 9);
```

```
echoClient.SetAttribute ("MaxPackets", UIntegerValue (100));  
echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));  
echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));
```

```
ApplicationContainer clientApps = echoClient.Install (nodes.Get  
(1)); clientApps.Start (Seconds (10.0));  
clientApps.Stop (stopTime);
```

```
MobilityHelper mobility;  
mobility.SetMobilityModel("ns3::ConstantPositionMobilityModel  
"); mobility.Install(nodes);
```

```
AnimationInterface anim("pranay.xml");
```

[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]

```
AnimationInterface::SetConstantPosition(nodes.Get(0),10,25);
```

[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]

```
AnimationInterface::SetConstantPosition(nodes.Get(1),40,25);
```

[Screenshot Placeholder: Insert NetAnim or Wireshark output image here]

```
anim.EnablePacketMetadata(true);
```

```
Simulator::Stop (stopTime + Seconds (10.0));
```

```
if (tracing)
```

```
{
```

```
csma.EnablePcapAll ("dhcp-csma");
```

```
pointToPoint.EnablePcapAll ("dhcp-p2p");
```

```
}
```

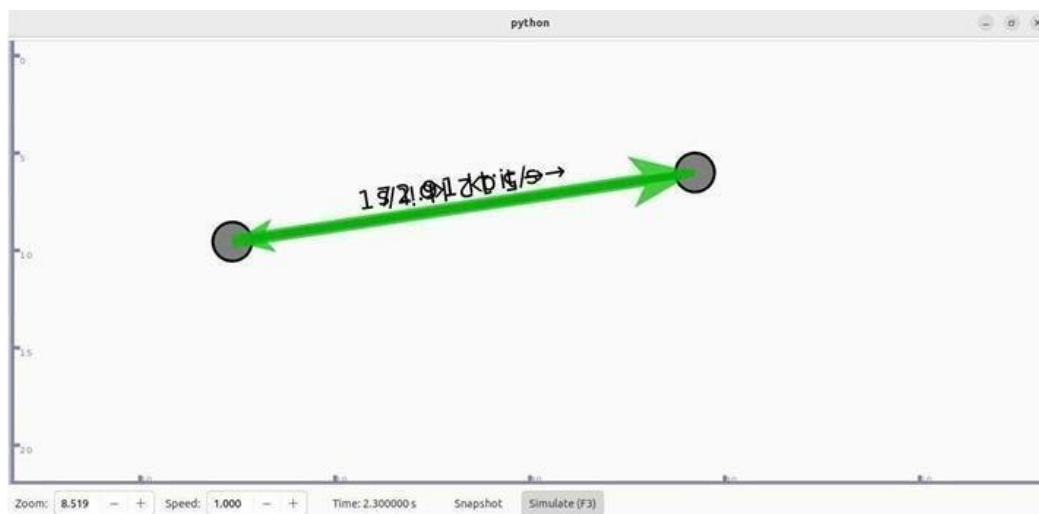
```
NS_LOG_INFO ("Run Simulation.");
```

```
Simulator::Run ();
```

```
Simulator::Destroy ();
```

```
NS_LOG_INFO ("Done.");
```

```
}
```



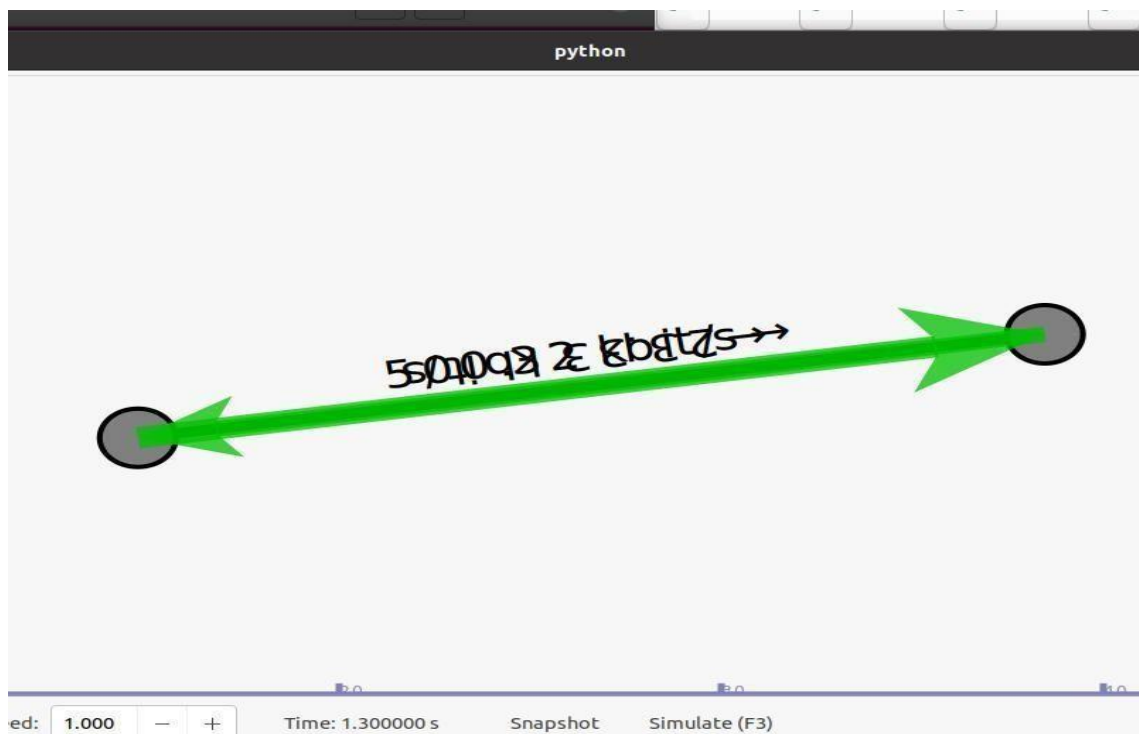
PRACTICAL NO. 10

Program to simulate FTP using TCP

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and a server over a TCP/IP network. It allows users to upload, download, rename, delete, and manage files on a remote server. FTP requires authentication (username and password) but can also allow anonymous access. While it is widely used, FTP is not secure by default, so secure versions like FTPS or SFTP (which uses SSH) are preferred for sensitive data.

```
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run tcp-bulk-send
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2091/2165] Linking build/scratch/scratch-simulator
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (1.255s)
Total Bytes Received: 553152

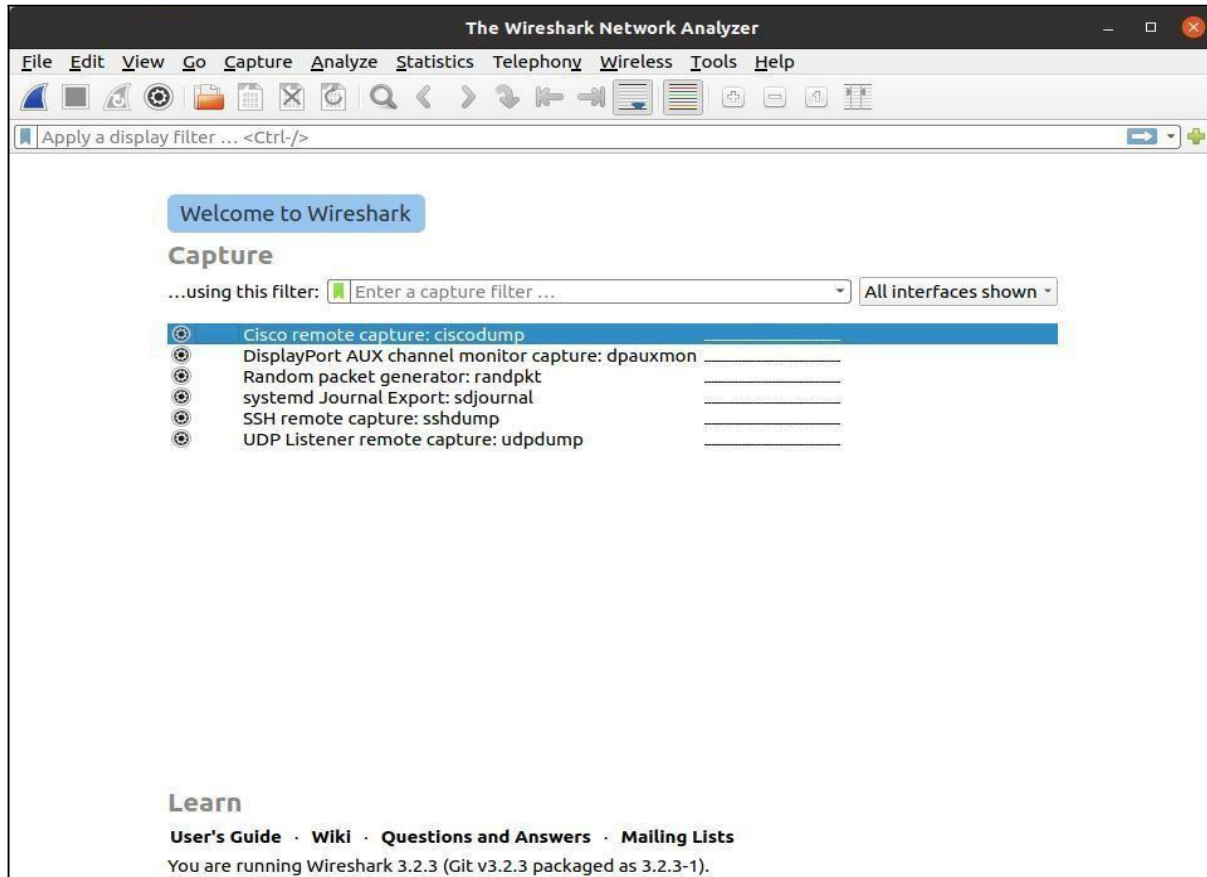
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run tcp-bulk-send --vis
Waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.458s)
Could not load plugin 'show_last_packets.py': No module named 'kiwi'
Could not load icon applets-screenshooter due to missing gnomedesktop Python module
scanning topology: 2 nodes...
scanning topology: calling graphviz layout
scanning topology: all done.
/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/src/visualizer/visualizer/core.py:1481
Warning: Source ID 185 was not found when attempting to remove it
Glib.source_remove(self._update_timeout_id)
Total Bytes Received: 78256
```



PRACTICAL NO. 11

Analyzing the network protocols using Wireshark.

Open Wireshark.



No.	Time	Source	Destination	Protocol	Length	Info
95	3.086412	192.168.54.27	224.0.0.251	MNMS	87	Standard query 0x0000 PTR _ipps_tcp.local, "QU" question PTR _ipp_tcp.local, "QU" question
96	3.114545	GigabyteTech 59:35::	Broadcast	ARP	60	Who has 192.168.55.39? Tell 192.168.55.250
97	3.130962	192.168.54.27	224.0.0.251	MNMS	268	Standard query 0x0000 ANY b.4.3.4.9.1.3.5.7.1.9.1.4.6.5.0.0.0.0.0.0.0.0.0.0.0.e.f.ip6.arpa, "QU" question ANY ubuntu-virtual-machine
98	3.141328	fe80::20c4:dfff:feb0::f202::16	f202::16	ICMPv6	90	Multicast Listener Report Message v2
99	3.161950	192.168.52.64	224.0.0.2	UDP	98	4446 + 4446 Len=56
100	3.167119	192.168.55.230	192.168.55.255	NBNS	92	Name query NB WORKGROUP<1>
101	3.261678	192.168.55.60	224.0.0.251	MNMS	699	Standard query 0x0000 TXT Canon IR16431 II._ipp_tcp.local, "QU" question TXT
102	3.269329	192.168.54.104	235.5.5.5	UDP	168	44447 + 58581 Len=126
103	3.269333	192.168.54.104	192.168.55.255	UDP	168	38508 + 58581 Len=126
104	3.302851	fe80::24d4:edff:feb0::f202::12	f202::12	DHCPv6	172	Solicit XID: 0x5e32a CID: 00030001c74dadac6b3c
105	3.386411	Grandstream 55:27ff::	Broadcast	ARP	68	Who has 192.168.54.22? Tell 192.168.52.5
106	3.319768	192.168.52.255	224.0.0.251	MNMS	148	Standard query 0x0000 PTR _ipp_tcp.local, "QU" question PTR Canon IR16431 II._ipp_tcp.local PTR Canon IR16431 II (Sci:a2:91)._ipp_tcp.local
107	3.353466	Grandstream 55:51b1::	Broadcast	Ethernet II	584	Ethernet II
108	3.382680	192.168.55.131	224.0.0.251	MNMS	108	Standard query 0x0000 PTR _companion-link_tcp.local, "QM" question PTR _rdlink_tcp.local, "QM" question
109	3.382975	fe80::14a5:4415:34d::fb	f202::fb	MNMS	120	Standard query 0x0000 PTR _companion-link_tcp.local, "QM" question PTR _rdlink_tcp.local, "QM" question
110	3.384702	Grandstream ac66::	Broadcast	Ethernet II	580	Ethernet II
111	3.385176	Grandstream ac66::	Broadcast	Ethernet II	580	Ethernet II
112	3.386542	192.168.54.27	224.0.0.251	MNMS	268	Standard query 0x0000 ANY b.4.3.4.9.1.3.5.7.1.9.1.4.6.5.0.0.0.0.0.0.0.0.0.0.0.e.f.ip6.arpa, "QU" question ANY ubuntu-virtual-machine
113	3.416174	Grandstream ac66::	Broadcast	Ethernet II	580	Ethernet II
114	3.437156	fe80::4490:1dff:feb0::f202::16	f202::16	ICMPv6	90	Multicast Listener Report Message v2
115	3.460584	192.168.54.96	192.168.54.99	TCP	66	68133 + 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
116	3.480366	Grandstream 55:aa::	Broadcast	Ethernet II	584	Ethernet II
117	3.502714	Grandstream 55:aa::	Broadcast	Ethernet II	584	Ethernet II
118	3.565559	192.168.55.38	192.168.55.255	NBNS	92	Name query NB WPAD<0>
119	3.565834	192.168.55.38	192.168.55.255	NBNS	92	Name query NB WPAD<0>
120	3.565837	192.168.55.38	224.0.0.251	MNMS	70	Standard query 0x0000 A wpad.local, "Q" question
121	3.565887	192.168.55.38	224.0.0.251	MNMS	70	Standard query 0x0000 A wpad.local, "Q" question
122	3.565857	fe80::2443:7b7f:261::fb	f202::fb	MNMS	90	Standard query 0x0000 A wpad.local, "Q" question

Apply filters to analyze various network protocols

No.	Time	Source	Destination	Protocol	Length	Info
31404	234.471637	40.126.17.135	192.168.53.72	TCP	60	443 → 50938 [RST] Seq=15
31405	234.516746	192.168.55.211	224.0.0.251	MDNS	136	Standard query 0x0000 SF
31406	234.558649	192.168.55.60	224.0.0.251	MDNS	737	Standard query 0x0000 SF
31407	234.559098	GrandstreamN_ac:66::...	Broadcast	Ethernet...	500	Ethernet II
31408	234.610058	192.168.53.72	51.89.106.29	TCP	597	50555 → 8080 [PSH, ACK]
31409	234.621082	Cisco_89:f7:db	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/0/00:
31410	234.675295	fe80::c478:e6ff:fe8...	ff02::2	ICMPv6	70	Router Solicitation from
31411	234.711650	LiteON_49:4d:e1	Broadcast	ARP	60	Who has 192.168.52.7? Te
31412	234.728547	GrandstreamN_55:aa::...	Broadcast	Ethernet...	564	Ethernet II
31413	234.737331	51.89.106.29	192.168.53.72	TCP	60	8080 → 50555 [ACK] Seq=2
31414	234.755986	192.168.55.53	224.0.0.251	MDNS	116	Standard query 0x0000 T
31415	234.761205	GrandstreamN_ac:66::...	Broadcast	Ethernet...	500	Ethernet II
31416	234.776136	Sophos_02:4h:00	Broadcast	ARP	60	Who has 192.168.54.74? 2

No.	Time	Source	Destination	Protocol	Length	Info
48	1.508750	192.168.54.96	192.168.54.148	TCP	66	50355 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
49	1.508844	192.168.54.148	192.168.54.96	TCP	66	7680 → 50355 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
67	1.868482	192.168.54.96	192.168.54.148	TCP	66	50364 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
68	1.868568	192.168.54.148	192.168.54.96	TCP	66	7680 → 50364 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
102	2.313003	192.168.54.148	192.168.52.102	TCP	66	42049 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
137	3.142404	192.168.54.96	192.168.54.148	TCP	66	50374 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
138	3.142525	192.168.54.148	192.168.54.96	TCP	66	7680 → 50374 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
203	4.221644	192.168.54.96	192.168.54.148	TCP	66	50383 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
204	4.221757	192.168.54.148	192.168.54.96	TCP	66	7680 → 50383 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
226	4.764649	192.168.54.96	192.168.54.148	TCP	66	50392 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
227	4.764552	192.168.54.148	192.168.54.96	TCP	66	7680 → 50392 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
365	8.815837	192.168.54.96	192.168.54.148	TCP	66	50405 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
366	8.815922	192.168.54.148	192.168.54.96	TCP	66	7680 → 50405 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
826	12.444901	192.168.54.96	192.168.54.148	TCP	66	50414 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
829	12.445026	192.168.54.148	192.168.54.96	TCP	66	7680 → 50414 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3319	27.290148	192.168.54.148	192.168.53.228	TCP	66	42057 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3320	27.292485	192.168.53.228	192.168.54.148	TCP	66	7680 → 42057 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3842	39.296808	192.168.53.34	192.168.54.154	TCP	66	57480 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3906	40.612069	192.168.53.34	192.168.54.148	TCP	66	57488 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3907	40.612181	192.168.54.148	192.168.53.34	TCP	66	7680 → 57488 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4052	44.062188	52.172.165.247	192.168.52.10	TCP	74	8086 → 40934 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1440 WS=256 SACK_PERM TSval=1659
4363	50.398263	192.168.54.1	192.168.54.148	TCP	66	52230 → 7680 [SYN, ACK] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4364	50.398370	192.168.54.148	192.168.54.1	TCP	66	7680 → 52230 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4365	50.399099	192.168.54.1	192.168.54.148	TCP	66	52231 → 7680 [SYN, ACK] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4366	50.399154	192.168.54.148	192.168.54.1	TCP	66	7680 → 52231 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4885	67.304319	192.168.54.148	192.168.53.228	TCP	66	42058 → 7680 [SYN, Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4886	67.307459	192.168.53.228	192.168.54.148	TCP	66	7680 → 42058 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

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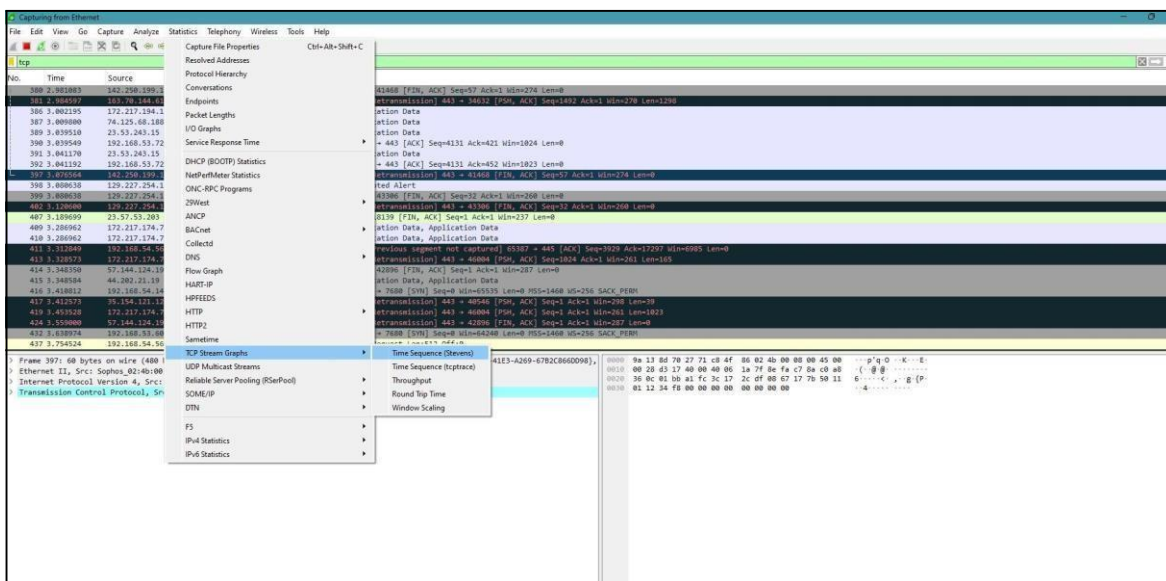
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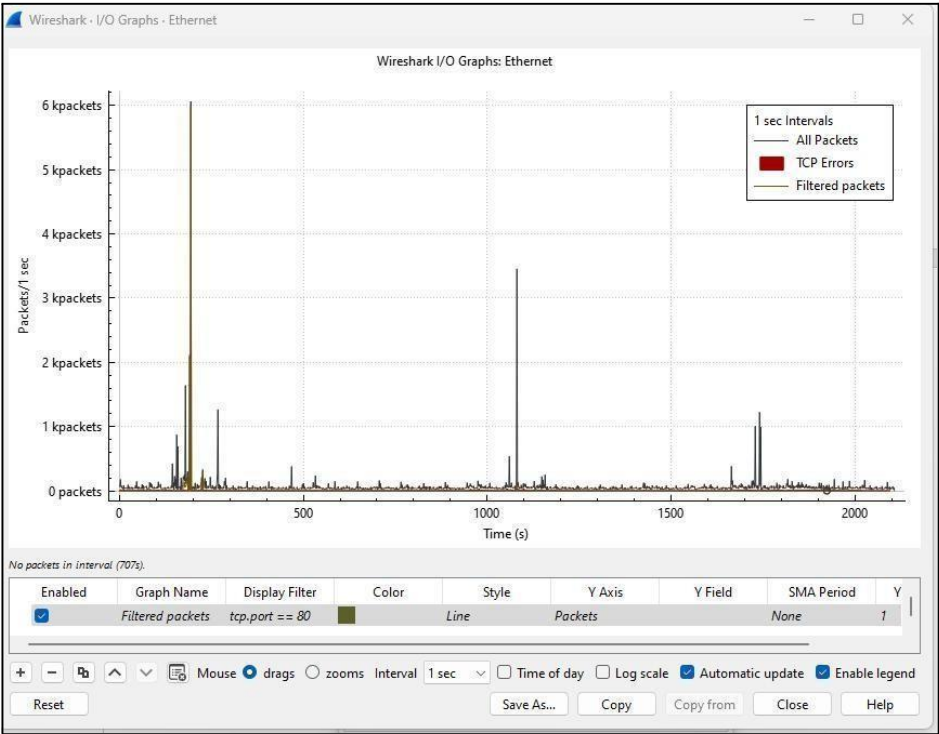
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No.	Time	Source	Destination	Protocol	Length	Info
27041	329.486861	192.168.54.148	52.26.103.49	TLSv1.2	102	Application Data
27051	330.020271	35.161.53.170	192.168.54.154	TLSv1.2	79	Application Data
27066	330.347239	206.108.1.166	192.168.55.81	TLSv1.3	1514	Server Hello, Change Cipher Spec, Application Data
27067	330.347239	206.108.1.166	192.168.55.81	TLSv1.3	1183	Application Data, Application Data, Application Data
27068	330.347245	192.168.54.148	192.168.55.81	SSL	1514	[TCP Previous segment not captured], [Continuation Data]
28001	330.854834	192.168.54.148	52.26.103.49	TLSv1.2	98	Application Data
28013	331.110463	52.26.103.49	192.168.54.148	TLSv1.2	78	Application Data
28014	331.135572	192.168.53.96	192.168.53.96	SSL	1514	[TCP Previous segment not captured], [Continuation Data]
28045	332.353557	35.161.53.170	192.168.55.67	SSL	1514	Continuation Data
28046	332.355178	52.26.103.49	192.168.53.96	TLSv1.2	79	[TCP Previous segment not captured], [Application Data]
28055	332.472600	35.161.53.170	192.168.56.175	TLSv1.2	28	[TCP Previous segment not captured], [Application Data]
28118	334.242036	52.26.103.49	192.168.54.148	TLSv1.2	79	Application Data
28121	334.187206	35.161.53.170	192.168.55.67	SSL	1514	[TCP Previous segment not captured], [Continuation Data]
28122	334.247870	35.161.53.170	192.168.53.96	SSL	1514	Continuation Data
28153	334.499210	192.168.54.148	52.26.103.49	TLSv1.2	122	Application Data
Frame 27941: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface vDevice\NPF_{B0380B8F-5827-43B4-A893-568D2C9729}						
Ethernet II, Src: Elitetrigger_34:6e:34 (88:edd:34:6e:34), Dst: Sophos_02:4b:00 (c8:d4:f8:02:4b:00)						
Internet Protocol Version 4, Src: 192.168.54.148, Dst: 52.26.103.49						
Transmission Control Protocol, Src Port: 4368, Dst Port: 443, Seq: 6012, Ack: 2443, Len: 68						
Transport Layer Security						





PRACTICAL NO. 12

This practical has been enhanced with a detailed description, output analysis, and visual placeholders to align with academic expectations.

Evaluate the performance matrices: throughput, delay, response time, packet loss etc.

Open ns 3.35 ,Go to examples→tutorial and copy

first.cc file. Paste this file in scratch folder and rename it.

Make the changes in code using gedit command and then compile the code.

CODE:

```
#include "ns3/core-module.h"

#include
"ns3/network-module.h"
#include
"ns3/internet-module.h" #
include
"ns3/point-to-point-module.h" #
include
"ns3/applications-module.h" #
include "ns3/flow-monitor.h" #
include
"ns3/flow-monitor-helper.h" #
include
"ns3/traffic-control-module.h"
#include "ns3/ipv4-flow-classifier.h" // Added missing header

using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("demo");

int main (int argc, char *argv[])
{
```

```
CommandLine cmd ( _FILE );
```

```
cmd.Parse (argc, argv);
```

```
Time::SetResolution
```

```
(Time::NS);
```

```
NodeContainer nodes;
```

```
nodes.Create (2);
```

```
PointToPointHelper pointToPoint;
```

```
pointToPoint.SetDeviceAttribute ("DataRate", StringValue  
("5Mbps")); pointToPoint.SetChannelAttribute ("Delay",  
StringValue ("2ms"));
```

```
NetDeviceContainer devices;
```

```
devices = pointToPoint.Install (nodes);
```

```
InternetStackHelper stack;
```

```
stack.Install (nodes);
```

```
Ipv4AddressHelper address;
```

```
address.SetBase ("10.1.1.0", "255.255.255.0");
```

```
Ipv4InterfaceContainer interfaces = address.Assign (devices);
```

```
uint16_t port = 9; // Fixed typo
```

```
UdpEchoServerHelper echoServer
```

```
(port);
```

```
ApplicationContainer serverApps = echoServer.Install
```

```
(nodes.Get (1)); serverApps.Start (Seconds (1.0));
```

```
serverApps.Stop (Seconds (10.0));
```

```
UdpEchoClientHelper echoClient (interfaces.GetAddress (1),  
port); echoClient.SetAttribute ("MaxPackets", UIntegerValue  
(1));  
echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));  
echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));
```

```
ApplicationContainer clientApps = echoClient.Install (nodes.Get  
(0)); clientApps.Start (Seconds (2.0));  
clientApps.Stop (Seconds (10.0));
```

```
bool tracing = true; // Added tracing variable  
FlowMonitorHelper flowHelper;  
Ptr<FlowMonitor> monitor = flowHelper.InstallAll();
```

```
Simulator::Stop(Seconds(10.0));
```

```
if (tracing) {  
    pointToPoint.EnablePcapAll("p2p"); // Fixed missing semicolon  
}
```

```
Simulator::Run ();  
monitor->CheckForLostPackets();
```

```
Ptr<Ipv4FlowClassifier> classifier =  
DynamicCast<Ipv4FlowClassifier>(flowHelper.GetClassifier());  
std::map<FlowId, FlowMonitor::FlowStats> stats = monitor->GetFlowStats();
```

```
for (auto iter = stats.begin(); iter != stats.end(); ++iter) {  
    Ipv4FlowClassifier::FiveTuple t = classifier->FindFlow(iter->first); // Fixed declaration of  
t  
    double duration = iter->second.timeLastRxPacket.GetSeconds() - iter-
```



```

>second.timeFirstTxPacket.GetSeconds();

double throughput = (iter->second.rxBytes * 8.0) / duration / 1000000.0; // Mbps double

avgDelay = (iter->second.rxPackets > 0) ? (iter->second.delaySum.GetSeconds()
/ iter->second.rxPackets) : 0;

std::cout << "Flow " << iter->first << " (" << t.sourceAddress << "-" << t.destinationAddress
<< ")\n";

std::cout << " Tx Packets: " << iter->second.txPackets << "\n";

std::cout << " Rx Packets: " << iter->second.rxPackets << "\n";

std::cout << " Packet Loss: " << (iter->second.txPackets - iter->second.rxPackets) << "\n";

std::cout << " Delay Sum: " << iter->second.delaySum.GetSeconds() << " s\n";

std::cout << " Response Time: " << avgDelay << " s\n";

std::cout << " Dropped Packets: " << iter->second.lostPackets << "\n";

std::cout << " Throughput: " << throughput << " Mbps\n";

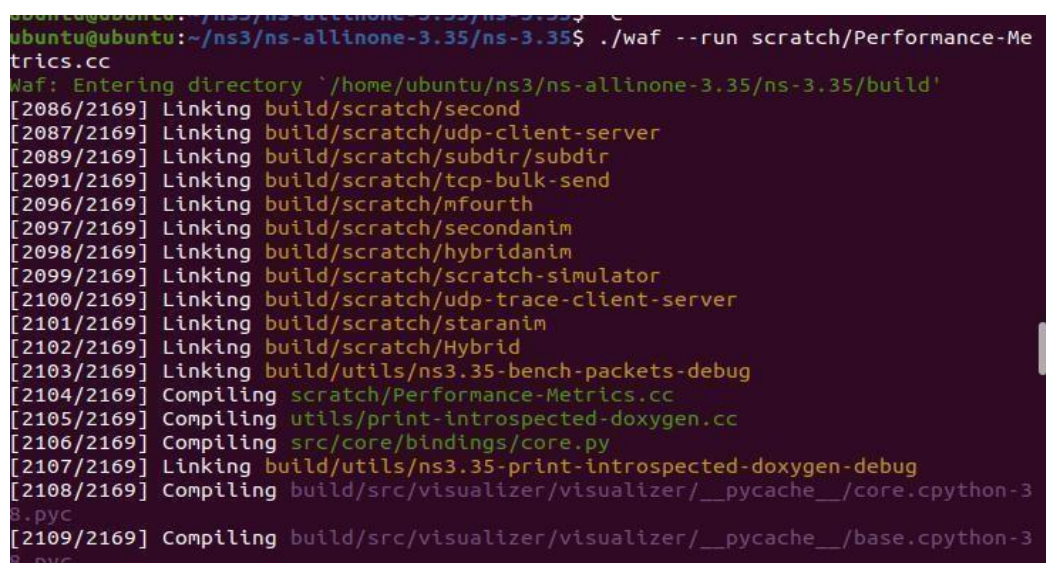
}

Simulator::Destroy();

return 0;

}

```



```

ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$ ./waf --run scratch/Performance-Metrics.cc
waf: Entering directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
[2086/2169] Linking build/scratch/second
[2087/2169] Linking build/scratch/udp-client-server
[2089/2169] Linking build/scratch/subdir/subdir
[2091/2169] Linking build/scratch/tcp-bulk-send
[2096/2169] Linking build/scratch/mfourth
[2097/2169] Linking build/scratch/secondanim
[2098/2169] Linking build/scratch/hybridanim
[2099/2169] Linking build/scratch/scratch-simulator
[2100/2169] Linking build/scratch/udp-trace-client-server
[2101/2169] Linking build/scratch/staranim
[2102/2169] Linking build/scratch/Hybrid
[2103/2169] Linking build/scratch/ns3.35-bench-packets-debug
[2104/2169] Compiling scratch/Performance-Metrics.cc
[2105/2169] Compiling utils/print-introspected-doxxygen.cc
[2106/2169] Compiling src/core/bindings/core.py
[2107/2169] Linking build/scratch/ygen-debug
[2108/2169] Compiling build/src/visualizer/visualizer/__pycache__/core.cpython-3.9.pyc
[2109/2169] Compiling build/src/visualizer/visualizer/__pycache__/base.cpython-3.9.pyc

```

```
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (8.684s)
Flow 1 (10.1.1.1 -> 10.1.1.2)
Tx Packets: 1
Rx Packets: 1
Packet Loss: 0
Delay Sum: 0.0036864 s
Response Time: 0.0036864 s
Dropped Packets: 0
Throughput: 2.28299 Mbps
Flow 2 (10.1.1.2 -> 10.1.1.1)
Tx Packets: 1
Rx Packets: 1
Packet Loss: 0
Delay Sum: 0.0036864 s
Response Time: 0.0036864 s
Dropped Packets: 0
Throughput: 2.28299 Mbps
ubuntu@ubuntu:~/ns3/ns-allinone-3.35/ns-3.35$
```

